KHWAJA FAREED UNIVERSITY OF ENGINEERING AND INFORMATION TECHNOLOGY, RAHIM YAR KHAN

Institute of Computer Science

Computer Science Module Handbook

Academic Session: 2019 – 2023

Program: BS Computer Science

This Module Handbook provides the detailed content and topic coverage of core Computer Science modules taught during the BS Computer Science program (Session 2019 - 2023) at the Institute of Computer Science, KFUEIT

KHWAJA FAREED UEIT RAHIM YAR KHAN | Institute of Computer Science

Programming Fundamentals + Lab (4 Credit hours)					
Credit Hours:	3 + 1	**Course Code**	COSC-1101 COSC-1201	**Instructor**	Ms. Sameen Fatima

Objectives:

This course provides a comprehensive introduction to the core concepts of high-level programming, using the C++ language to focus on algorithmic thinking, structured programming, and the fundamental syntax required for advanced application development

Course Learning Outcomes:

- Students will be able to design, implement, and debug basic programs in C++ using structured programming principles.
- Students will demonstrate proficiency in fundamental programming constructs including data types, control flow structures, and functions.
- Students will be able to analyze simple problems and develop efficient algorithms using arrays and strings to manage data.

Course Contents:

- **Programming Fundamentals:** Problem-solving concepts, algorithms, pseudocode, flowcharts, program structure, syntax, compilation process, and basic C++ programming environment.
- **Data Types and Variables:** Primitive data types, constants, variable declaration and initialization, type casting, and input/output operations.
- **Operators and Expressions:** Arithmetic, relational, logical, assignment, increment/decrement, conditional and other fundamental C++ operators.
- **Control Flow:** Selection structures (if, if-else, nested if, switch) and repetition structures (while, do-while, for), including loop control statements.
- **Functions and Modularization:** Function declaration/definition, parameters and arguments, return values, pass-by-value, pass-by-reference, function overloading, scope rules, and modular program design.
- **Recursion:** Recursive functions, base and recursive cases, and applications of recursion.
- **Arrays and Strings:** One-dimensional and multidimensional arrays, array processing and passing arrays to functions, C-style strings, string manipulation and basic string functions.
- **Pointers and Memory Management:** Pointers, pointer arithmetic, addresses, references, dynamic memory allocation and deallocation.
- **Structures and Basic User-Defined Types:** Structures, records, and organization of related data.
- **File Handling:** File input/output, reading from and writing to files, and basic data persistence.
- **Problem Solving and Algorithms:** Basic algorithm implementation, searching, sorting, simple computational problems, and introduction to algorithm efficiency.
- **Complexity and Runtime Analysis:** Basic time and space complexity concepts and analysis of simple algorithms.
- **Debugging and Testing:** Syntax, logical and runtime errors, debugging techniques, tracing, testing, validation, and program correctness.
- **Introduction to Classes and Objects:** Basic classes and objects as a foundation for subsequent OOP.

Text Books:
1. C++ How to Program by *Paul Deitel & Harvey Deitel*.
Recommended Books:
1. C++ Primer by *Stanley B. Lippman* et al

Sr. No	Weekly	Topics
1	Week 1st	Introduction to Programming Paradigms, C++ Environment Setup, Compilation vs. Interpretation, Algorithms, Pseudocode, Flowcharts, Basic Program Structure
2	Week 2nd	C++ Syntax, Variables, Constants, Basic Data Types (int, float, char, bool), Input/Output and Type Casting
3	Week 3rd	Arithmetic, Relational, Logical Operators, Assignment, Increment/Decrement, Conditional Operators and Expressions
4	Week 4th	Selection Structures I: if, if-else, and Nested if-else Statements
5	Week 5th	Selection Structures II: switch, Statement and Conditional Operator
6	Week 6th	Repetition Structures I: while, do-while, for Loops
7	Week 7th	Repetition Structures II: Nested Loops, Loop Control (break, continue)
8	Week 8th	Functions I: Defining and Calling Functions, Prototypes, Libraries, Parameters and Return Values
9	Week 9th	Functions II: Pass by Value, Pass by Reference, Function Overloading, Recursion, Scope Rules
10	Week 10th	Arrays I: One-Dimensional Arrays, Array Processing, Searching and Basic Sorting
11	Week 11th	Arrays II: Multidimensional Arrays, Passing Arrays to Functions, Complexity and Runtime Analysis
12	Week 12th	Strings: C-style Strings, String Manipulation, Basic String Functions
13	Week 13th	Pointers: Addresses, Pointer Arithmetic, References, Dynamic Memory Allocation
14	Week 14th	Structures and User-Defined Data Types, Basic File Handling and File I/O
15	Week 15th	File Handling (Input/Output with Files), Debugging, Testing, Error Handling and Program Validation
16	Week 16th	Introduction to Classes and Objects (Basic OOP Concepts as a Foundation for Next Course)

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Object Oriented Programming					
Credit Hours:	3	**Course Code**	COSC-1102	**Instructor**	Mr. Muiz-u-din

Objectives:

This course introduces the fundamental concepts of object-oriented programming using Java. It focuses on designing software using classes and objects and applying core OOP principles such as encapsulation, inheritance, polymorphism, and abstraction.

Course Learning Outcomes:

- Students will be able to design and implement programs using classes and objects in Java.
- Students will demonstrate understanding of core OOP concepts including encapsulation, inheritance, polymorphism, and abstraction.
- Students will be able to use constructors, methods, and access modifiers effectively.
- Students will develop modular, reusable, and efficient programs using Java.
- Students will apply exception handling and file handling in Java applications.

Course Contents:

- Introduction to OOP and Java basics: JVM, syntax, modular program design, and program structure
- Classes and Objects: Defining classes, creating objects, access modifiers
- Methods and Constructors: Method overloading, constructors
- Encapsulation and Abstraction: Data hiding and interfaces
- Inheritance: Types of inheritance, code reusability, and reusable components
- Polymorphism: Method overloading and method overriding
- Interfaces and Abstract Classes: Abstraction and implementation of abstract behavior
- Introduction to functional programming and functional languages; higher-order functions, functional data structures, static and dynamic typing.
- Exception Handling: try-catch, finally, throw/throws
- File Handling: Reading and writing files in Java
- Arrays and Strings in Java
- Introductory monads, and functional-logic programming

Text Books:

1. **Java:** How to Program by *Deitel & Deitel*
2. Head First Java by *Kathy Sierra & Bert Bates*

Recommended Books:

1. Effective Java by Joshua Bloch

Weekly Plan

Sr. No	Weekly	Topics
1	Week 1st	Introduction to Java and OOP concepts
2	Week 2nd	Java syntax, variables, and data types
3	Week 3rd	Classes and objects
4	Week 4th	Methods and constructors
5	Week 5th	Encapsulation and access modifiers
6	Week 6th	Inheritance basics
7	Week 7th	Types of inheritance and implementation
8	Week 8th	Polymorphism (overloading & overriding)
9	Week 9th	Abstract classes and interfaces
10	Week 10th	Exception handling
11	Week 11th	Arrays and strings
12	Week 12th	File handling, Packages, Java libraries, modular program design, and reusable components
13	Week 13th	Introduce functional programming as a contrasting programming paradigm compare imperative / object-oriented and functional styles, and demonstrate higher-order functions.
14	Week 14th	GUI basics, cover functional data structures and static versus dynamic typing; connect the concepts to reusable components where appropriate.
15	Week 15th	Project development, use a short guided practical example to introduce program transformations, monads, and functional-logic programming at survey level.
16	Week 16th	Final revision and evaluation

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Course title: Object Oriented Programming-Lab					
Credit Hours	1	**Course Code**	COSC-1202	**Instructor**	Mr. Muizzud din

Objectives:

This module focused on developing practical OOP proficiency by implementing advanced programming concepts (Encapsulation, Inheritance, Polymorphism) in Java, crucial for building modular and scalable software systems.

Sr. No	Weekly	Topics
1	Week 01-02	Java Environment Setup (JDK, IDE). Writing and compiling basic Java programs, understanding the main method
2	Week 03-04	Classes and Objects Creation: Practical Implementation of Methods, Constructors, and Object Interaction
3	Week 05	Encapsulation: Data protection using access modifiers (private, public) and getter/setter methods.
4	Week 06	Arrays of Objects and Object initialization: Complex data structures handling
5	Week 07-08	Creating sub-classes and super-classes, understanding method overriding.
6	Week 09	Polymorphism: Run-time and compile-time polymorphism (Method Overloading).
7	Week 10	Abstract Classes and Interfaces implementation: Designing templates for class behavior.
8	Week 11-12	Exception Handling (Try-Catch-Finally)
9	Week 13	Packages, Modular Program Design, File Input/Output (I/O) using Java streams
10	Week 14	Reusable Components, Collections Framework basics: ArrayLists and HashMaps
11	Week 15-16	Comprehensive OOP Project: Practical software implementation of a small system (Student record system) using Inheritance, Polymorphism, and Encapsulation

Data Structures & Algorithms + Lab (4 Credit hours)					
Credit Hours	3 + 1	**Course Code**	COSC-2101 COSC-2201	**Instructor**	Mr. Saqib Ubaid

Objectives:

This module focused on algorithmic problem solving, practical implementation and empirical analysis of complex data structures and fundamental algorithms in Java. Emphasis was placed on solving advanced computational problems, particularly in graph theory (Shortest Path, MST), and comparing algorithm efficiency.

Course Learning Outcomes (CLOs)

CLO 1:
Students will be able to explain and apply fundamental data structures, abstract data types, and algorithmic concepts, including arrays, linked lists, stacks, queues, trees, graphs, hash tables, determinism, termination, and asymptotic complexity.

CLO 2:
Students will be able to design and implement fundamental searching, sorting, tree, hashing, and graph algorithms, including BFS, DFS, Dijkstra's algorithm, and Minimum Spanning Tree algorithms, using appropriate data structures.

CLO 3:
Students will be able to analyze algorithm correctness, time and space complexity, runtime efficiency, and select suitable data structures and algorithms for practical computational problem-solving.

Course Content

Abstract Data Types (ADTs), arrays, linked lists, stacks, queues, recursion, searching and sorting algorithms, binary trees, BST and AVL trees, heaps and priority queues, hash tables and collision resolution, graph representations and traversal algorithms (BFS, DFS). Advanced graph-algorithmic problems: matching, graph-coloring algorithms, and flow algorithms, including problem modelling, algorithm selection, and implementation with suitable data structures. Shortest path and Minimum Spanning Tree algorithms (Dijkstra, Kruskal, Prim), algorithmic problem solving, determinism and termination, and basic time and space complexity, implementation, correctness, and runtime analysis. Elementary algebraic algorithms: Euclid's algorithm, the Chinese remainder theorem, discrete Fourier transform (DFT), and matrix multiplication, with attention to implementation and algorithmic efficiency.

Sr. No	Weekly	Topics
1	Week 01-02	Practical implementation of Abstract Data Types (ADTs) using Arrays/Vectors; basic data structure concepts, algorithm characteristics, determinism and termination, asymptotic analysis (Big-O, Big-Theta, Big-Omega), time and space complexity
2	Week 03	Implementation of Linked Lists (Singly, Doubly, and Circular): Node creation, insertion, deletion, traversal, searching, and dynamic memory management
3	Week 04	Implementation of Stacks and Queues using Arrays and Linked Lists; Circular Queues, basic Queue variations, applications including expression evaluation, recursion and function-call management
4	Week 05	Sorting Algorithms (1 of 6): Practical implementation of Bubble Sort and Selection Sort and comparative analysis

5	Week 06	Sorting Algorithms: Merge Sort, Quick Sort and Heap Sort; recursive and divide-and-conquer approaches, complexity analysis and comparison of sorting algorithms
6	Week 07	Advanced Sorting and Recursion: Merge Sort and Quick Sort; recursion, divide-and-conquer strategy, recurrence concepts, correctness and runtime analysis
7	Week 08	Trees: Binary Trees, tree terminology, node creation, recursive tree traversal (Preorder, Inorder, Postorder), level-order traversal and basic tree operations
Mid Term Exam		
8	Week 09	Binary Search Trees (BSTs) and AVL Trees: Search, insertion, deletion, tree traversal, balancing, rotations and height analysis
9	Week 10	Graph Data Structures: Directed and undirected graphs, weighted and unweighted graphs, vertices and edges, degree, paths and cycles, adjacency matrix and adjacency list representations. Introduce matching, graph coloring, and flow problems as advanced graph applications.
10	Week 11	Graph Traversal Algorithms: Breadth-First Search (BFS) and Depth-First Search (DFS), traversal applications, connectivity and path finding
11	Week 12	Shortest Path Algorithms: Dijkstra's Algorithm, weighted graphs, min-heap and priority queue implementation, runtime analysis and applications. Matching, coloring, and flow algorithms.
12	Week 13	Minimum Spanning Trees: Kruskal's Algorithm, Union-Find/Disjoint Set data structure, cycle detection, weighted graphs and runtime analysis
13	Week 14	Minimum Spanning Trees: Prim's Algorithm, priority queues/heaps, comparison with Kruskal's Algorithm, correctness and complexity analysis
14	Week 15	Hashing and Hash Tables: Hash functions, hash table operations, collision resolution techniques (chaining and open addressing), performance and load factor analysis. Euclid’s algorithm and the Chinese remainder theorem, followed by the discrete Fourier transform (DFT) and matrix multiplication as examples of algebraic algorithms. Compare their computational costs.
15	Week 16	Final Lab Project: Algorithmic problem solving, data structure selection, algorithm implementation, debugging/testing, correctness and runtime analysis. Implement or analyze one advanced graph or algebraic algorithm, including correctness, testing, and runtime analysis.
Final Term Exam		

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Tentative Weekly Course Plan

Course Title	Design & Analysis of Algorithms	**Instructor**	Mr. Mutiullah Jamil
Course Code	COSC-3115	**Credit Hour**	3

Objective:

This course develops a strong foundation in the design, analysis, implementation, and evaluation of algorithms for computational problems. It covers asymptotic analysis, recurrence relations, major algorithmic paradigms, graph algorithms, optimization techniques, computational complexity, and practical performance evaluation. Students learn to design correct and efficient algorithms, analyze their time and space requirements, and select appropriate approaches for complex computational and data-driven problems.

Course Learning Outcomes:

CLO 1: Analysis: Mathematically analyze the time and space complexity of algorithms using asymptotic notation, recurrence relations, and appropriate efficiency measures.
CLO 2: Design: Design algorithms using major paradigms including Divide-and-Conquer, Greedy, Dynamic Programming, Backtracking, Branch-and-Bound, and Local Search to solve computational and optimization problems.
CLO 3: Implementation: Implement and evaluate fundamental and advanced algorithms, including sorting, searching, graph, shortest-path, and spanning-tree algorithms, while ensuring correctness and practical efficiency.
CLO 4: Complexity: Analyze computational problems using concepts of tractability, NP, NP-hardness, NP-completeness, reductions, approximation algorithms, and the limitations of efficient algorithmic solutions.

Course Contents

- Algorithm design and analysis fundamentals, asymptotic notation and time/space complexity, recurrence relations and Master Theorem, algorithm correctness and proof techniques, searching and sorting algorithms, Divide-and-Conquer, Greedy algorithms, Dynamic Programming and optimization, Backtracking, Branch-and-Bound and Local Search, graph algorithms including BFS, DFS, topological sorting, shortest paths and Minimum Spanning Trees, hashing, B-Trees and disjoint-set structures, NP-completeness and reductions, approximation algorithms, and practical implementation, correctness, efficiency, and performance analysis

Approximation ratios and guarantees for Travelling Salesman, Bin Packing, and Knapsack; negative approximation results, the gap technique, Probabilistically Checkable Proofs (PCP), and the PCP theorem. Continuous and discrete optimization, the Bellman principle, linear and integer optimization, duality, cutting planes, and branch-and-bound. Robust and stochastic optimization, probabilistic constraints, multistage programs, and constrained optimization modelling. Network-flow optimization, optimal-flow formulations, and comparison of network optimization objectives beyond shortest paths and minimum spanning trees. Introduction to GAMS and AMPL for formulating and exploring optimization models.

Reference Books

- *Introduction to Algorithms by Thomas H. Cormen, Charles E. Leiserson, Ronald L. Rivest, and Clifford Stein (CLRS).*
- *Algorithm Design by Jon Kleinberg and Éva Tardos.*

Sr. No	Weekly	Topics
1	Week 01-02	Algorithm definition, Role of algorithms in computing, Algorithm design and analysis, Growth of Functions, Asymptotic Notations and complexity, correctness and efficiency analysis
2	Week 03	Methods for solving recurrence relations: Substitution Method, Recursion Tree, and the Master Theorem
3	Week 04-05	Analysis and implementation of Insertion Sort, Merge Sort (Divide and Conquer), and Heapsort (using Priority Queues)
4	Week 06	Quicksort, Randomized Algorithms, and establishing the lower bounds for sorting
5	Week 07-08	Introduction to Greedy Strategy, Proof of Correctness, Activities Selection Problem, Huffman Coding, Minimum Spanning Trees (Prim's & Kruskal's)
6	Week 09	Principles of Dynamic Programming, optimization techniques, Optimal substructure, overlapping subproblems, the Bellman principle, and its application to dynamic optimization.
7	Week 10	Graph Representations, Graph Traversal Techniques: Breadth-First Search (BFS) and Depth-First Search (DFS), Topological Sort
8	Week 11	Dijkstra's Algorithm, Bellman-Ford Algorithm, and its application in negative weight cycles
9	Week 12	Network-flow optimization, and optimal-flow formulations. The Floyd-Warshall Algorithm
10	Week 13	Advanced data structures for efficiency: B-Trees (overview), Hash Tables, and Disjoint Sets
11	Week 14	Complexity classes P and NP, NP-completeness, reductions, approximation limits, negative approximation results, and the gap technique.
12	Week 15	Introduction to Approximation Algorithms (Vertex Cover), Local Search, Backtracking, Branch-and-Bound, and other appropriate algorithms relevant to Data Science. Approximation ratios and guarantees for selected Travelling Salesman, Bin Packing, and Knapsack problems; PCP and the PCP theorem.
13	Week 16	Linear and integer optimization, duality, cutting planes, robust and stochastic optimization, probabilistic constraints, multistage programs, and introductory applications of GAMS and AMPL.

Student Evaluation criteria:

Student Evaluation criteria:	5%
Surprise Test, Quizzes:	5%
Semester Project:	15%
Mid Term Paper:	25%
Final Term paper (Marks Distribution: 20 % from Mid Term and 80% from Final Term), i.e. 10 + 40=50:	50%
Total:	100%

Student Responsibilities:

A student must have attended at least 75% of the classes held in a course in order to be allowed to sit in the final examination. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments. Students are also responsible for doing all assigned work on time. Students may prepare Sketchbook for taking notes and for references.

Khwaja Fareed University of Engineering & Information Technology

Rahim Yar Khan

Institute of Computer and Software Engineering

Course Outline

COSC-1104 Software Engineering (3 Credit Hours)

Pre-Requisite: N/A
Program: Computer Science
Total Lectures: 32
Total Tutorials: 0

Instructor: Mr. Afzal Hussain
Office: Faculty Hall, ICT Building, KFUEIT
Email: *afzal.hussain@kfueit.edu.pk*
Office Hours: 08:00am -12:30 pm (Wednesday)

Course Introduction

This course emphasizes understanding the principles, methodologies, and practices of modern software engineering. The objective of this course is to provide students with in-depth knowledge of software development lifecycles (SDLC, Agile/Scrum, Waterfall), requirements analysis, software design and architecture, object-oriented design, implementation and testing, debugging, version control, software quality, documentation and collaborative development. Students will gain practical skills to design, model, evaluate, and maintain high-quality software systems.

Course Contents

1. **Introduction to Software Engineering & Lifecycles:**
   The Concept of Software Engineering, Professional Ethics, Software development lifecycles (SDLC, Agile/Scrum, Waterfall), Incremental and Spiral Models.
2. **Requirements Engineering:**
   Requirements Elicitation, requirements analysis, Functional vs Non-Functional Requirements, Software Requirement Specification (SRS) Development.
3. **Software Architecture & Design:**
   Architectural Patterns, software design and architecture, Component Level Design, Architectural Styles, Interface Design.
4. **Object-Oriented Design & System Modeling:**
   object-oriented design Principles, Structural and Behavioral UML Modeling (Class Diagrams, Use Case Diagrams, Sequence Diagrams, Activity Diagrams).
5. **Implementation, Testing & Debugging:**
   Coding Standards, implementation and testing Strategies, Unit Testing, Integration Testing, System Testing, Black-Box and White-Box Testing, debugging Techniques.
6. **Version Control & Collaborative Development:**
   Source Code Management, Git and GitHub Workflows, Branching Strategies, version control, Code Reviews, collaborative development Practices.
7. **Software Quality Assurance:**
   Concepts of Verification and Validation, software quality Attributes, SQA Planning, Software Metrics, Code Inspections and Audits.
8. **Documentation & Maintenance:**
   Technical documentation, User Manuals, Software Maintenance, Legacy System Evolution, Re-Engineering.
9. **Ethical and societal** considerations in software development, including inclusive design, decision-making under uncertainty, technology risks, and moral dilemmas, examined through software engineering case studies.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
CLO_1	Understand software development lifecycles (SDLC, Agile/Scrum, Waterfall) and foundational software engineering principles.	PLO1	C2 (Understanding)
CLO_2	Perform systematic requirements analysis, system modeling, and draft software documentation.	PLO2	C3 (Applying)
CLO_3	Apply principles of software design and architecture and object-oriented design using standard UML diagrams.	PLO3	C3 (Applying)
CLO_4	Execute implementation and testing, debugging, version control, SQA for software quality, and engage in collaborative development.	PLO5	C4 (Analyzing)

Overall Grading Policy

Assessment Items	Percentage
Quizzes	8%
Assignments	7%
Semester Project	10%
Midterm Exam	25%
Final Exam	50%

Text and Reference Books

- Ian Sommerville: *Software Engineering*
- Roger S. Pressman: *Software Engineering: A Practitioner's Approach*
- Craig Larman: *Applying UML and Patterns: An Introduction to Object-Oriented Analysis and Design*
- Martin Fowler: *UML Distilled: A Brief Guide to the Standard Object Modeling Language*

Administrative Instructions

- According to institute policy, 80% attendance is *mandatory* to appear in the final examination.
- Project proposal, SRS, architecture design, and final report must be submitted as per announced format and within due date.
- In any case, there will be no retake of (scheduled/surprise) quizzes.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Computer/LMS Usage

- Students will be required to use StarUML / Enterprise Architect for system modeling, Git & GitHub for version control, Visual Studio Code / IDEs for implementation, and Microsoft Word/PowerPoint for software documentation and project presentations

Lecture Breakdown

Week	Topics
Week 1	• Introduction to Software Engineering

	• Evolution of Software Engineering & The Software Crisis • Ethical and societal consequences of software design and deployment, illustrated through a case study.
Week 2	• Overview of Software development lifecycles (SDLC, Agile/Scrum, Waterfall) • Traditional Process Models: Waterfall, Prototyping, Spiral Model
Week 3	• Modern Agile Methodologies: Agile/Scrum Framework • Scrum Roles, Sprints, User Stories, and collaborative development
Week 4	• Requirements Engineering Process • Functional vs Non-Functional Requirements • Techniques for requirements analysis & Elicitation
Week 5	• Writing Software Requirement Specification (SRS) • Validation and Verification of Requirements • Requirements Traceability & Software documentation
Week 6	• Fundamentals of software design and architecture • Architectural Styles: Layered, Client-Server, MVC, Microservices
Week 7	• Object-oriented design Principles & Concepts • SOLID Design Principles (Single Responsibility, Open-Closed, etc.) • Inclusive design considerations in architectural and interface decisions, accounting for the needs of diverse users.
Week 8	• System Modeling with Unified Modeling Language (UML) • Structural Diagrams: Class Diagrams & Component Diagrams
Week 9	Mid Term Exam
Week 10	• Behavioral UML Diagrams: Use Case, Sequence, and Activity Diagrams • Mapping Requirements to Design Artifacts
Week 11	• Implementation and testing Strategies • Coding Standards and Refactoring • Version control with Git/GitHub and Branching Workflows
Week 12	• System Defect Tracking • Practical debugging Techniques & Strategies
Week 13	• Software Testing Levels: Unit, Integration, System, Acceptance Testing • Black-Box vs White-Box Testing Techniques
Week 14	• Software Quality Assurance (SQA) Concepts • Managing software quality Attributes and Software Metrics • Technology-risk identification, decision-making under uncertainty, trade-offs, and moral dilemmas in software-quality case studies.
Week 15	• Software Technical documentation Standards • Software Maintenance Types: Corrective, Adaptive, Perfective
Week 16	• collaborative development Project Submissions • Semester Project Code Reviews & Demos
Week 17	• Final Project Presentations & Evaluations
Week 18	Final Exam

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Tentative Weekly Course Plan

Course Title	Database Systems + Lab (4 Credit hours)		
Course Code	COSC-2103 and COSC-2203	**Credit Hour**	3 + 1
Name	Mr. Adeel Abid	**Email**	*adeel.abid@kfueit.edu.pk*

Objective

This course is to provide students with a comprehensive understanding of the principles, design, and management of Database Management Systems (DBMS). The course focuses on fundamental data models, especially the Relational Model, query languages, and the principles of database integrity and concurrency

Course Learning Outcomes (CLOs)

CLO 1: Data Modeling: Design and develop robust and normalized database schemas using the Entity-Relationship (E-R) Model and map them to the Relational Model.

CLO 2: Querying & Manipulation: Write complex and efficient queries using Structured Query Language (SQL), including joins, sub-queries, and data manipulation commands (DML, DDL).

CLO 3: Normalization: Apply Normalization Theory (up to 3NF and BCNF) to eliminate data redundancy and anomalies, ensuring data integrity and consistency.

CLO 4: Transaction & Control: Explain the concepts of database transactions, concurrency control (e.g., locking), recovery mechanisms, and the basic principles of database security.

Course Content

Database architecture and ANSI-SPARC architecture, data independence, database design and ER/EER modeling, ER-to-relational mapping, relational model and relational algebra, relational calculus, keys and integrity constraints, SQL including DDL, DML, joins, nested queries, aggregate functions, views, stored procedures, functions and triggers, functional dependencies and normalization from 1NF to BCNF, lossless decomposition, query processing and optimization, transaction management and ACID properties, concurrency control and deadlocks, recovery techniques, physical storage and file organization, indexing using B-Trees/B+ Trees, hierarchical, network and object-oriented database models, NoSQL and data warehousing concepts, database security and authorization, and practical database implementation and SQL laboratory exercises.

	1. Reference Books	
	i	*Database System Concepts by Abraham Silberschatz, Henry F. Korth, and S. Sudarshan*
	ii	*Fundamentals of Database Systems by Ramez Elmasri and Shamkant B. Navathe*
	2. Web Resources	
	i.	w3school.com
	ii.	geeksforgeeks.org/dbms

Week #	Topics (outline of main topics and sub topics)
1 - 2	Database Architecture, ANSI-SPARC Architecture, Data Independence, Database Users and DBMS vs File Systems
3 - 4	Relational Model Concepts, Keys (Primary, Foreign, Candidate), Integrity Constraints, Relational Algebra Operations, and Relational Calculus
5 - 6	Entity-Relationship (ER) Modeling, ER Components (Entities, Attributes, Relationships), Enhanced ER (EER) Modeling, and ER-to-Relational Schema Mapping
7 - 8	SQL DDL/DML, Joins, Aggregate Functions, GROUP BY/HAVING, Nested Queries, Views
Mid Term Exam	
9 - 10	Advanced SQL, Set Operations, Stored Procedures, Functions and Triggers
11 - 12	Functional Dependencies, Normalization Forms (1NF, 2NF, 3NF, BCNF), Lossless Join Decomposition
13	Concept of Transactions, ACID Properties (Atomicity, Consistency, Isolation, Durability), Transaction States
14	Concurrency Control, Lock-Based Protocols, Deadlocks and Timestamp Ordering
15	Physical Storage, File Organization, Indexing, B-Trees / B+ Trees, Recovery and Backup
16	Hierarchical, Network and Object-Oriented Database Models, NoSQL, Data Warehousing, Database Security and Final Database Lab Project
Final Term Exam	

Assignments	5%
Surprise Test, Quizzes	5%
Presentation	15%
Mid Term Paper	25%
Final Term paper (***Marks Distribution:*** 20 % from Mid Term and 80% from Final Term), i.e. 10 + 40=50	50%
Total	**100%**

Student Responsibilities:

A student must have attended at least **75%** of the classes held in a course in order to be allowed to sit in the final examination. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments. Students are also responsible for doing all assigned work on time. Students may prepare Sketchbook for taking notes and for references.

Department of Computer Science

Tentative Weekly Course Plan

Course Title	Web Programming		
Course Code	COSC-4120	**Credit Hour**	3

Course Learning Outcomes (CLOs):

1. Understand Web technologies and client-server architecture along with the core mechanics of HTTP.
2. Develop front-end user interfaces with HTML/CSS and JavaScript for interactive dynamic web programming.
3. Implement server-side processing, forms and data handling, and persistent database integration.
4. Build, configure, and execute the end-to-end deployment of web-based applications.

Resource Material	**1. Reference Books**	
	i	Jon Duckett: *HTML and CSS: Design and Build Websites*
	ii	Jon Duckett: *JavaScript and JQuery: Interactive Front-End Web Development*
	iii	Robin Nixon: *Learning PHP, MySQL & JavaScript*
	2. Web Resources	
	i.	https://www.w3schools.com/
	ii.	https://www.freecodecamp.org/news/learn-web-development-for-free/

TOPICS TO BE COVERED

Week #	Topics (outline of main topics and sub topics)	Tutorial /Laboratory
1	• Introduction to Web technologies and client-server architecture • Evolution of the web and modern standards	
2	• Mechanics of HTTP request and response flow • Client-side vs server-side roles	Quiz
3	• Document structure and semantic elements using HTML/CSS • Standard web page formatting ➢ **Assignment to achieve CLO 1**	
4	• Styling pages and responsive grid design with HTML/CSS • Media queries and page layouts	
5	• Introduction to dynamic web programming using JavaScript • Basic syntax, variables, and control flow	

6	• Interactive DOM manipulation with JavaScript • Dynamic element generation	
7	• Event handling and client-side page interaction • Input field monitoring ➢ **Assignment to achieve CLO 2**	
8	• Building web user interface forms and data handling • Client-side data validation	Quiz
Mid Term Exam		
10	• Fundamentals of server-side processing logic • Setting up backend execution environments	
11	• Routing requests and server-side processing workflows • Session and cookie management	
12	• Database integration principles for web applications • Connecting web servers to persistent storage ➢ **Assignment to achieve CLO 3**	
13	• Executing database operations and queries • Processing backend form data into databases	Quiz
14	• Web security fundamentals, input sanitization, and session safety • Application protection basics	
15	• Full-stack web application development consolidation • Preparation for deployment of web-based applications ➢ **Assignment to achieve CLO 4**	
16	• Live web application project testing and reviews	Quiz
17	• Final project presentations	
	Final Term Exam	

Student Evaluation criteria:

Workshop / Assignments/Case study	5%
Surprise Test, Quizzes	5%
Semester Project	15%
Mid Term Paper	25%
Final Term paper	50%
Total	**100%**

Department of Computer Science

Instructor:

Mr. Maqsood Ahmad

Email: maqsood.ahmad@kfueit.edu.pk

Consulting hours: 12:00pm-02:00pm Monday-Thursday.

Office: 2.03L, Department of Computer Science, ICT Building, KFUEIT

Office Hours (Instructor):
Monday - Thursday: 08:00 – 16:30

Student Responsibilities:

Students must attend class. Failure to attend class may result in failure in the course. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments, students are responsible for knowing what they say. Students are also responsible for doing all assigned work on time. Excessive absences (more than 03) will result in "F Grade". Students may prepare Sketchbook for taking notes and for references.

Instructor/ Subject Specialist

Chairman/ Head of Department

Khawaja Fareed University of Engineering & IT
Rahim Yar Khan

Tentative Course Plan

Department of Computer Science

Class: **Semester:** **Session:**

Instructor:	Ms. Humaira Anwer	**Email:** humaira.anwer@kfueit.edu.pk	
Course Title:	Artificial Intelligence + Lab	**Program:**	BSCS
Course Number:	COSC-3112 and COSC-3212	**Credit Hours:**	(3+1) = 4

Lecture Time:		
Course Objective & Outcomes: 1. Understand key components in the field of artificial intelligence 2. Analyze artificial intelligence techniques for practical problem solving 3. Install/Establish Implement classical artificial intelligence techniques		
Methods of Teaching: • Class participation is very important and will be evaluated on the bases of class attendance, punctuality, ability to contribute in class discussions, participation in answer questions session and preparation level with assigned readings. All students are expected to prepare for and participate in class. • The methodology of Direct / Live teaching at class room as per Lecture schedule where student have the freedom to interact and ask any question to teacher regarding the current Lecture topic. • Student would also have option to contact with teacher at specific office hours (may change according to summer and winter season) regarding the course related queries.		
Resource Material	**1. Reference Book**	
	i	Artificial Intelligence. A Modern Approach, Stuart Russell and Peter Norvig, 4th Edition, Prentice Hall, Inc., 2020; ISBN-10: 0134610997
	ii	Artificial Intelligence with Python, Your Complete Guide to building Intelligent apps using Python 3.x and TensorFlow 2, Alberto Artasanchez, Prateek Joshi 2nd Edition, 2020 ISBN-10: 178646439X
	2. Web Resources	
	i.	https://www.ailab.com.au/resources/online-resources/
	ii.	https://deeplearning.mit.edu/
Office Help Hours	During Official Hours	

SEQUENCE OF TOPICS TO BE COVERED		
Week	**Topics** (outline of main topics and sub topics)	**Book / Tutorial**
1	**Introduction to Artificial Intelligence:** • What Is AI? • The Foundations of Artificial Intelligence • The History of Artificial Intelligence • The State of the Art • Risks and Benefits of AI	**Chapter No. 1**
2	**Intelligent Agents:** • Agents and Environments • Good Behavior: The Concept of Rationality • The Nature of Environments • The Structure of Agents • Perception-action cycle. • Environmental observations and action selection.	**Chapter No. 2**
3	**Problem Solving By Searching:** • Problem-Solving Agents • Example Problems • Search Algorithms • Uninformed Search Strategies	**Chapter No. 3**
4	**Problem Solving By Searching:** • Uninformed Search Strategies • Informed (Heuristic) Search Strategies • Heuristic Functions	**Chapter No. 3**
5	**Adversarial Search and Games:** • Game Theory • Optimal Decisions in Games • Heuristic Alpha--Beta Tree Search • Monte Carlo Tree Search • Stochastic Games • Partially Observable Games • Limitations of Game Search Algorithms	**Chapter No. 5**
6	**Constraint Satisfaction Problems:** • Defining Constraint Satisfaction Problems • Constraint Propagation: Inference in CSPs • Backtracking Search for CSPs • Local Search for CSPs The Structure of Problems	**Chapter No. 6**
7	**Logical Agents:** • Knowledge-Based Agents • The Wumpus World - Explanation • Logic • Propositional Logic: A Very Simple Logic • Propositional Theorem Proving • Effective Propositional Model Checking • Agents Based on Propositional Logic	**Chapter No. 7**

8	**First Order Logic & Knowledge Representation:** • Representation Revisited • Syntax and Semantics of First-Order Logic • Using First-Order Logic • Knowledge Engineering in First-Order Logic • Ontologies and Knowledge Graphs	**Chapter No. 8**
9	**Inference in First Order Logic & Knowledge Systems:** • Propositional vs First-Order Inference • Unification and First-Order Inference • Forward Chaining • Backward Chaining • Rule-based and Expert Systems • Resolution & Automated Theorem Proving	**Chapter No. 9**
10	**Learning:** **Automated Planning & Multiagent Decision Making:** • Automated Planning • Properties of Multiagent Environments • Non-Cooperative Game Theory • Cooperative Game Theory • Making Collective Decisions	**Chapter No. 18**
11	**Discrete Data Analysis & Learning from Examples:** • Discrete Data Analysis • Forms of Learning • Supervised Learning • Learning Decision Trees • Model Selection and Optimization • The Theory of Learning • Inductive programming from examples • Comparison with other learning approaches.	**Chapter No. 19**
12	**Learning from Examples:** • Linear Regression and Classification • Nonparametric Models • Ensemble Learning • Developing Machine Learning Systems • AI/ML method selection for specific problems. • Assumptions, suitability, and limitations of selected methods.	**Chapter No. 19**
13	**Deep Learning:** • Simple Feedforward Networks • Computation Graphs for Deep Learning • Convolutional Networks • Learning Algorithms • Generalization • Recurrent Neural Networks • Unsupervised Learning and Transfer Learning • Computer-vision features and visual perception models. • Relationship between visual perception and CNN-based deep learning • Applications	**Chapter No. 21**

14	**Natural Language Processing:** • Language Models and Grammar • Parsing and Augmented Grammars • Complications of Real Natural Language • Natural Language Tasks	**Chapter No. 23**
15	**Deep Learning for Natural Language Processing:** • Limitations of Feature-Based NLP Models • Word Embeddings • Recurrent Neural Networks • Sequence-to-sequence Models • The Transformer Architecture Pretraining and Transfer Learning	**Chapter No. 24**
16	**Reinforcement Learning** • Learning from Rewards • Passive Reinforcement Learning • Active Reinforcement Learning • Safe Exploration • Generalization in Reinforcement Learning • Policy Search • Applications of Reinforcement Learning	**Chapter No. 22**
	Final Term Exam	

Student Evaluation criteria:

Workshop / Assignments/Case study	5%
Surprise Test/Sudden Test , Quizzes	5%
Semester Project	15%
Mid Term Paper	25%
Final Term paper	50%
Total	**100%**

Student Responsibilities:

Students must attend class. Failure to attend class may result in failure in the course. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beepers must be Turned off (Proper classroom decorum [behavior] adopted. Course outlines and calendars explain requirements and assignments, students are responsible for knowing what they say. Students are also responsible for doing all assigned work on time. Excessive absences (more than 03) will result in "F Grade". Students may prepare Sketchbook for taking notes and for references.

Instructor/Tutor

Approved by:

Chairman/ HOD/ Subject Specialist

KHWAJA FAREED UEIT RAHIM YAR KHAN | Institute of Computer Science

Course Title: Data Science					
Credit Hours:	3	**Course Code**	COSC-3109	**Instructor**	Mr. Maqsood Ahmad

Objectives:

This course introduces the systematic process of Extracting Knowledge from Data. It emphasizes the foundational steps of the Data Science workflow, including data acquisition, cleaning, statistical analysis, and visualization. The course provides a strong statistical and computational background for handling and interpreting complex datasets, serving as a necessary precursor to advanced Machine Learning studies.

Course Learning Outcomes:

- **CLO 1: Data Acquisition and Wrangling:** Implement practical techniques for data integration, cleaning, and transformation using programmatic tools (e.g., Python/R libraries).
- **CLO 2: Statistical Foundation:** Apply fundamental descriptive and inferential statistics (e.g., probability distributions, hypothesis testing) to explore and summarize data effectively.
- **CLO 3: Data Visualization:** Select and create effective data visualizations (e.g., histograms, scatter plots, heatmaps) to uncover patterns and communicate insights.
- **CLO 4: Predictive Basics:** Understand and implement basic linear modeling concepts and evaluate the quality of simple predictive models.

Text Books:

1. Doing Data Science by *Cathy O'Neil and Rachel Schutt*
2. R for Data Science by *Hadley Wickham and Garrett Grolemund*

Web resources:

i.	https://kaggle.com/learn
ii.	https://jakevdp.github.io/PythonDataScienceHandbook
iii.	https://scikit-learn.org

Weekly Plan of "Data Science" by Mr. Maqsood Ahmad

Sr. No	Weekly	Topics
1	Week 1-2	• Data Science pipeline overview, Role of statistics, Data sources and types, data acquisition and preprocessing, Introduction to data manipulation tools with Python (NumPy, Pandas) and R
2	Week 3-4	• Data cleaning and missing-value handling (Imputation), Outlier detection, feature engineering (Categorical Encoding, One-Hot Encoding), scaling and normalization, Data integration techniques
3	Week 5-6	• Exploratory data analysis (EDA), statistical summaries (Mean, Median, Variance), Data summarization, Measures of central tendency and dispersion
4	Week 7-8	• Probability Distributions (Normal, Binomial), Central Limit Theorem, Confidence Intervals, Introduction to Hypothesis Testing
Mid Term Exam		
5	Week 9-10	• Principles of effective visualization using Python (Matplotlib, Seaborn), Univariate and Bivariate visualizations, plotting time series data, Creating dashboards (basic)
6	Week 11-12	• Review of relational data concepts, Basic SQL queries for data extraction and joinin tables, Connecting to data sources
7	Week 13-14	• Introduction to Machine Learning with Scikit-Learn: Simple Linear Regression, Clustering (K-Means), Dimensionality Reduction (PCA), Model Evaluation & Validation Metrics (MSE, R^2 Score, Confusion Matrix, Silhouette Score) • Comparison of supervised, unsupervised, and reinforcement learning.
8	Week 15	• Task formulation and the role of feedback. • Introduction to concepts like Bayes' theorem, Basic decision rules (not algorithms like k-NN/SVM)
9	Week 16	• Applying the entire workflow (from raw data to final report) on a real-world dataset • Documentation of data/model selection, evaluation methods, and results • Reporting limitations, risks, and responsible-use considerations.
Final Term Exam		

Khawaja Fareed University of Engineering & IT
Rahim Yar Khan

Tentative Course Plan

Department of Computer Science

Class: **Semester:** **Session:**

Instructor:	Mr. Shahzad Hussain	**Email:** shahzad.hussain@kfueit.edu.pk	
Course Title:	Machine Learning	**Program:**	BSCS
Course Number:	COSC-3107	**Credit Hours:**	3

Lecture Time:		
Course Objective & Outcomes: 1. To introduce students to the basic concepts and techniques of Machine Learning. 2. To develop skills of using recent machine learning software for solving practical problems. 3. To gain experience of doing independent study and research.		
Methods of Teaching: · Class participation is very important and will be evaluated on the bases of class attendance, punctuality, ability to contribute in class discussions, participation in answer questions session and preparation level with assigned readings. All students are expected to prepare for and participate in class. · The methodology of Direct / Live teaching at class room as per Lecture schedule where student have the freedom to interact and ask any question to teacher regarding the current Lecture topic. · Student would also have option to contact with teacher at specific office hours (may change according to summer and winter season) regarding the course related queries.		
Resource Material	**1. Recommended & Reference Books**	
	i	Fundamentals of Machine Learning for Predictive Data Analytics, Algorithms, worked examples and case studies 2020, ISBN-10: 0262029448
	ii	Practical Machine Learning with Python, 2017; ISBN-10: 1484232062
	iii	Introduction to Machine Learning with Python, 2018 ISBN-10: 1449369413
	2. Web Resources	
	i.	https://machinelearningmastery.com/start-here/
	ii.	https://medium.com/analytics-vidhya
Office Help Hours	During Official Hours	

SEQUENCE OF TOPICS TO BE COVERED		
Week	**Topics (outline of main topics and sub topics)**	**Book / Tutorial**
1	**Machine Learning and Applications:** **Introduction to Machine Learning and Applications:** Why you should learn machine learning with us; who this specialization is for and what you will be able to do; getting started with the tools for the course; getting started with Python and Jupyter notebook; getting started with SFrames for data engineering and analysis	
2	**Machine Learning Methods** Supervised Learning: Classification; Regression. Un-supervised Learning: Clustering, Dimensionality reduction, Anomaly detection, Association rule-mining; Semi-Supervised Learning and Reinforcement Learning. Batch Learning, Online Learning, Instance Based Learning and Model Base Learning.	
3	**Machine Learning and Real-world Applications (Use-Case Classification):** Analyzing the Sentiment of Reviews; what is intelligent restaurant Review System; Examples of Classification tasks; Linear classifiers; Decision Boundary **Evaluating of Classification:** Training and evaluating a classifier; model evaluation (Cross-validation, Confusion Matrix, Precision, Recall, ROC-AUC); What a good accuracy; Fale Positive, false negative; Learning curves; Class prediction. Model limitations, risks, and appropriate use alongside classification metrics. **Assignment:** **To implement various Supervised Learning Machine Learning Models on same dataset and for the same task.**	
4	**Machine Learning and Real-world Applications (Use-Case Regression):** Predicting a house price; what good and how might you naively address it? A linear regression: a model-based approach; adding higher order effects. **Evaluating regression models:** Model limitations, risks, and appropriate use alongside regression metrics. Evaluating overfitting via training / test split; Training / test curves; adding their features; Other regression examples **Assignment:** **To implement various Supervised Machine Learning Regression Models on same dataset and for this task.**	
5	**The CRISP-DM Process Model** Business Understanding, Define Business Problem, Assess and Analysis Scenarios, Define the data-mining problem, Project plan, Data Understanding, Data Collection, Data Description, Exploratory Data Analysis, Data Quality Analysis, Data Preparation, Data Integration, Data Wrangling, Attribute Generation and Selection **Implementation CRISP-DM Process Model in real world problem: Real-World Case Study: Predicting Student Grant Recommendations**	

6	**Processing, Wrangling and Visualizing Data:** **Data Collection:** CSV, JSON, XML, HTML and Scraping, HTML, Web Scraping, SQL, **Data Description:** Numeric, Text, Categorical **Data Wrangling:** Understanding Data, Filtering Data, Typecasting, Transformations, Imputing Missing Values, Handling Duplications, Handling Categorical Data, Normalizing Values, String	
	Manipulations.	
7	**Processing, Wrangling and Visualizing Data:** **Data Summarization:** **Data Visualization:** Visualizing with Pandas, Line Charts, Bar Plots, Histograms, Pie Charts, Box Plots, Scatter Plots, Visualization with Matplotlib, Figures and Subplots, Plot formatting, Legends, Axis Controls, Annotations, Global Parameters, Python Visualization Ecosystem.	
8	**Data Preprocessing & Feature Engineering:** Data and Datasets, Features, Models; Feature Engineering on Numeric Data, Feature Engineering on Categorical Data. Feature Engineering on Text Data, Feature Engineering on Temporal Data, Feature Engineering on Image Data	
Mid Exam		
9	**Dimensionality Reduction (PCA & Feature Selection):** Standardized Scaling, Min-Max Scaling, Robust Scaling, Feature Selection, Threshold-Based Methods, Statistical Methods, Recursive Feature Elimination, Model-Based Selection, **Dimensionality Reduction:** Feature Extraction with Principal Component Analysis,	
10	**Building, Tuning and Deploying Models:** **Supervised Learning:** Classification and Regression; Generalization, Overfitting and Underfitting; Supervised Machine Learning Algorithms; **k-Nearest Neighbors Model**: k-Neighbors classification, analyzing KneighboursClassifier, k-neighbors regression, analyzing k-nearest neighbors region, strength, weaknesses and parameters **Linear Models**: Linear Models: Linear Model for regression and classification; Linear/Logistic Regression; Ridge regression; Lasso; regularization; Linear models for multi-class classification, strength, weaknesses and parameters.	
11	**Building, Tuning and Deploying Models:** **Supervised Learning:** **Decision Trees:** Building Decision Trees; Controlling complexity of Decision Trees & Probabilistic Models: Naive Bayes classification; Building Decision Trees; Controlling complexity of Decision Trees; Analyzing Decision Trees; Feature Importance in trees; Strength, weaknesses and parameters, Ensembles of Decision Trees; Random Forests and Gradient Boosted Regression Trees (GBM). **Kernelized Support Vector Machines:** Linear Models and Non-linear features, the kernel trick, understanding SVMs, tuning SVM parameters, Preprocessing data for SVMs, strength and weakness.	

12	**Building, Tuning and Deploying Models:** **Supervised Learning:** **Decision Trees:** Building Decision Trees; Controlling complexity of Decision Trees & Probabilistic Models: Naive Bayes classification; Building Decision Trees; Controlling complexity of Decision Trees; Analyzing Decision Trees; Feature Importance in trees; Strength, weaknesses and parameters, Ensembles of Decision Trees; Random Forests and Gradient Boosted Regression Trees (GBM). **Kernelized Support Vector Machines:** Linear Models and Non-linear features, the kernel trick, understanding SVMs, tuning SVM parameters, Preprocessing data for SVMs, strength, weakness and parameters	
13	**Building, Tuning and Deploying Models:** **Supervised Learning:** **Neural Networks (Deep Learning):** the neural network model, tuning neural networks, strength, weaknesses and parameters **Classifying movie reviews:** a binary classification example **Neural-network/model-training unit:** Gradient-based loss minimization and a comparative introduction to gradient, conjugate-gradient, Newton / quasi-Newton methods and line search **Advanced Optimization Tutorial:** Constrained model fitting and KKT conditions, Penalty, barrier, and interior-point methods.	

	Building, Tuning and Deploying Models: **Un-Supervised Learning:** Types of unsupervised learning; Preprocessing and Scaling; Dimensionality Reduction, Feature Extraction and Manifold Learning, PCA, NMF, Manifold learning with t-SNE	.5
14	**Building, Tuning and Deploying Models:** **Un-Supervised Learning:** **Clustering:** k-Means clustering, agglomerative clustering, DBSCAN, comparing and evaluating clustering algorithms; evaluating clustering with ground truth.	
15	**Real World Case Studies:** **Analyzing Bike sharing Trends** The Bike sharing Dataset; Problem Statement; Exploratory and Data Analysis; Regression Analysis; Modeling; Testing; Next Steps	
16	**Real World Case Studies:** **Galaxy Classification** Evaluation results, model limitations, risks, and responsible use, Business Understanding, Data Understanding, Data Preparation, Modeling, Evaluation, Deployment	
	Final Term Exam	

Student Evaluation criteria:

Workshop / Assignments/Case study	5%
Surprise Test/Sudden Test , Quizzes	5%
Semester Project	15%
Mid Term Paper	25%
Final Term paper	50%
Total	**100%**

Student Responsibilities:

Students must attend class. Failure to attend class may result in failure in the course. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beepers must be Turned off (Proper classroom decorum [behavior] adopted. Course outlines and calendars explain requirements and assignments; students are responsible for knowing what they say. Students are also responsible for doing all assigned work on time. Excessive absences (more than 03) will result in "F Grade". Students may prepare Sketchbook for taking notes and for references.

Department of Computer Science

Tentative Weekly Course Plan

Course Title	Introduction to NLP		
Course Code	COSC-3121	**Credit Hour**	3

Course Learning Outcomes (CLOs):

1. Understand core concepts, text preprocessing methods, and language modeling techniques in Natural Language Processing.
2. Analyze and implement algorithms for POS tagging, syntactic parsing, and word embeddings.
3. Design and evaluate deep learning and statistical models for text classification and sentiment analysis.

Resource Material	**1. Reference Books**	
	i	Daniel Jurafsky & James H. Martin: *Speech and Language Processing* (3rd Edition Draft)
	ii	Christopher D. Manning & Hinrich Schütze: *Foundations of Statistical Natural Language Processing*
	iii	Steven Bird, Ewan Klein, & Edward Loper: *Natural Language Processing with Python (NLTK)* (O'Reilly)
	2. Web Resources	
	i.	https://www.nltk.org/book/
	ii.	https://web.stanford.edu/class/cs224n/

TOPICS TO BE COVERED

Week #	Topics (outline of main topics and sub topics)	Tutorial /Laboratory
1	• Introduction to NLP • Language Thoughts	
2	• Regular Expression and Automata • Text Normalization & Preprocessing (Tokenization, Stemming, Lemmatization)	Quiz
3	• Minimum Edit Distance • Backtrace for Computing Alignments ➢ **Assignment to achieve CLO 1**	
4	• N-gram Language Models • Evaluation and Perplexity	
5	• Lexical Classes & POS	

	• POS Tagger and Practice	
6	• Text Representation & Feature Weighting • Term Frequency-Inverse Document Frequency (TF-IDF)	
7	• Distributed Representation & Word Embeddings • Word2Vec (CBOW, Skip-Gram) and GloVe **Assignment to achieve CLO 2**	
8	• Text Classification Pipelines • Sentiment Analysis Foundations & Practice • Quiz 1	Quiz
Mid Term Exam		
10	• Deep Learning for Text Processing • Recurrent Neural Networks (RNN/LSTM) for Text	
11	• Constituency Grammars • Context-Free Grammars (CFG) & Sentence Construction Rules • Syntactic Parsing (CFG)	
12	• Sentence Construction Rules & Treebank • CKY Parsing Algorithm ➢ **Assignment to achieve CLO 3**	
13	• Probabilistic Context-Free Grammars (PCFGs) • Probabilistic CKY Algorithm	Quiz
14	• Logical Representation of Sentence Meaning • First-Order Logic for Computational Semantics	
15	• Syntax-driven Semantic Analysis • Semantic Parsing & Meaning Extraction **Assignment to achieve CLO 4**	
16	• Sequence-to-Sequence Architecture & Attention Mechanisms • Modern NLP Applications & Hands-on Framework Practice	Quiz
17	• Semester Project Presentations & Review	
	Final Term Exam	

Student Evaluation criteria:

Workshop / Assignments/Case study	5%
Surprise Test, Quizzes	5%
Semester Project	15%
Mid Term Paper	25%

Final Term paper	50%
Total	**100%**

Instructor:

Ms. Humaira Anwer

Email: humaira.anwer@kfueit.edu.pk

Office: Faculty Hall, Institute of Computer Science, ICT Building, KFUEIT

Office Hours (Instructor):
Monday - Thursday: 11:00 – 13:30

Student Responsibilities:

Students must attend class. Failure to attend class may result in failure in the course. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments, students are responsible for knowing what they say. Students are also responsible for doing all assigned work on time. Excessive absences (more than 03) will result in "F Grade". Students may prepare Sketchbook for taking notes and for references.

Instructor/ Subject Specialist

Chairman/ Head of Department

KHWAJA FAREED UEIT RAHIM YAR KHAN | Institute of Computer Science

Institute of Computer Science

Course Outline

COSC-4121 Deep Learning and Applications (3 Credit Hours)

Pre-Requisite: Machine Learning
Program: Computer Science
Total Lectures: 32
Total Tutorials: 0

Instructor: Mr. Shahzad Hussain
Office: 3.2R, ICT Building, KFUEIT
Email: *shahzad.hussain@kfueit.edu.pk*
Office Hours: 12:00pm -1:30 pm (Wednesday)

Course Introduction

This course provides a comprehensive foundation in deep learning, covering fundamental neural network architectures, optimization strategies, and practical applications. Students will explore Artificial Neural Networks (Perceptron, MLP), Convolutional Neural Networks (CNNs) for Computer Vision, Recurrent Neural Networks (RNN/LSTM) for Sequential Data, Attention Mechanisms, Transfer Learning, and Generative Deep Learning. Hands-on framework practice using PyTorch and TensorFlow is integrated throughout the course to prepare students for real-world artificial intelligence tasks.

Course Contents

1. **Introduction to Artificial Neural Networks & Fundamentals:**
   Artificial Neural Networks (Perceptron, MLP), Activation functions (ReLU, Sigmoid, Softmax, Tanh), Binary Classification, Multi-class Classification, PyTorch/TensorFlow framework practice.
2. **Model Training, Optimization & Regularization:**
   Backpropagation algorithm, Gradient computation, Advanced optimization methods (SGD, RMSprop, Adam), Advanced regularization techniques (L1/L2 Weight Decay, Dropout).
3. **Foundations of Convolutional Neural Networks (CNNs) for Vision:**
   Convolutional Neural Networks (CNNs) for vision, Convnets vs. Densely Connected Networks (MNIST), Core convolution operations (Kernels, feature response maps), Padding (valid vs. same), Strides, Max-Pooling operation.
4. **Small Dataset Training & Data Augmentation:**
   Training a Convnet from scratch on small datasets (Dogs vs. Cats classification), Data preprocessing, Image scaling, Directory management, Automated batch processing using ImageDataGenerator, Overfitting diagnosis, Data Augmentation (Rotation, Shift, Shear, Zoom, Flip).
5. **Models & Transfer Learning:**
   Transfer Learning principles, Feature Extraction using pretrained networks (VGG16 convolutional base), Freezing layers, Attaching custom dense classifiers, Fine-tuning pretrained CNN models, Architectural overview (VGG16, ResNet50, Inception V3, Xception).
6. **Recurrent Neural Networks & Attention Mechanisms:**
   Recurrent Neural Networks (RNN/LSTM) for sequential data, Long-term dependencies, Vanishing gradients, Attention mechanisms, Practical sequential data modeling using PyTorch/TensorFlow.
7. **Generative Deep Learning & DeepDream:**
   Introduction to Generative Deep Learning, DeepDream algorithm (Gradient ascent on feature activations), Visualizing network representations, Multi-scale processing (Octaves) using pretrained Inception V3.

8. **Computer-vision applications:** Low-level perception, 2D image features, colour, and motion features.
9. **Application Case Studies:** Vision-based deep learning applications, including material diagnostics, model selection, suitability, and limitations.
10. **Project Reporting and Ethics:** Model selection, training, evaluation, results, and limitations. Ethical and social implications of applications affecting people or decisions.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
CLO_1	Understand the core concepts, mathematical principles, and architectures of deep learning models.	PLO1	C2 (Understanding)
CLO_2	Analyze and design deep learning solutions for computer vision and sequence modeling applications.	PLO2	C4 (Analyzing)
CLO_3	Develop, train, and evaluate end-to-end deep learning projects using PyTorch/TensorFlow frameworks.	PLO3	C6 (Creating)

Overall Grading Policy

Assessment Items	Percentage
Quizzes	5%
Assignments	5%
Semester Project	15%
Midterm Exam	25%
Final Exam	50%

Text and Reference Books

- Ian Goodfellow, Yoshua Bengio, Aaron Courville: *Deep Learning* (MIT Press)
- François Chollet: *Deep Learning with Python* (Manning Publications)
- Aurélien Géron: *Hands-On Machine Learning with Scikit-Learn, Keras, and TensorFlow* (O'Reilly)

Administrative Instructions

- According to institute policy, 80% attendance is *mandatory* to appear in the final examination.
- Project proposal, SRS, architecture design, and final report must be submitted as per announced format and within due date.
- In any case, there will be no retake of (scheduled/surprise) quizzes.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Computer/LMS Usage

- Students will be required to use Python, PyTorch / TensorFlow, Google Colab, Jupyter Notebooks, and MS Office (Word/PowerPoint) for assignments and semester project submissions.

Lecture Breakdown

Week	Topics
Week 1	• Artificial Neural Networks (Perceptron, MLP)

	• Activation functions (ReLU, Sigmoid, Tanh, Softmax) • Introduction to deep learning workflows
Week 2	• Binary Classification and Multi-class Classification foundations • Loss functions (MSE, Cross-Entropy) • PyTorch/TensorFlow framework practice setup
Week 3	• Backpropagation algorithm and computational graphs • Chain rule and manual gradient computation • Tensor operations and automatic differentiation
Week 4	• Advanced optimization techniques (SGD, Momentum, RMSprop, Adam) • Hyperparameter tuning and learning rate schedulers
Week 5	• Advanced regularization strategies (L1/L2 Weight Decay, Dropout) • Diagnosing overfitting vs. underfitting
Week 6	• Convolutional Neural Networks (CNNs) for vision • Densely Connected Networks vs. Convnets (MNIST Handwritten Digit Classification) • Low-level image / color features and learned convolutional feature maps.
Week 7	• Core convolution operations: Kernels, feature response maps, and output channels • Border effects, Padding (valid vs. same), and Convolution Strides
Week 8	• Max-Pooling operation and feature map downsampling • PyTorch/TensorFlow framework practice for CNNs • Quiz 1
Week 9	Mid Term Exam
Week 10	• Training a Convnet from scratch on small datasets (Dogs vs. Cats classification) • Data preprocessing, image scaling, and directory management • Automated batch processing using ImageDataGenerator
Week 11	• Overfitting diagnosis using training and validation accuracy/loss plots • Mitigating overfitting using Data Augmentation (Rotation, Shift, Shear, Zoom, Flip) and Dropout
Week 12	• Transfer Learning principles and generic feature representations • Feature Extraction using pretrained networks (VGG16 convolutional base) • Motion features and visual-application case studies, including material diagnostics where appropriate.
Week 13	• Freezing layers and attaching custom dense classifiers • Fine-tuning pretrained CNN models on small custom datasets • Architectural overview: VGG16, ResNet50, Inception V3, Xception
Week 14	• Recurrent Neural Networks (RNN/LSTM) for sequential data • Handling long-term dependencies and vanishing gradients in LSTMs
Week 15	• Attention mechanisms and sequence modeling • DeepDream algorithm: Gradient ascent on feature activations • Multi-scale processing (Octaves) and detail reinjection using pretrained Inception V3
Week 16	• Model selection, training, evaluation, results, and limitations. • Ethical and social implications where relevant.
Week 17	• Semester Project Presentations and Final Review
Week 18	Final Exam

Khwaja Fareed University of Engineering & Information Technology

PROPOSED TEACHING PLAN

Spring Semester 2022

Department of Computer Science

Teacher's name	Saima Khosa	**Email ID**	Saima.khosa@kfueit.edu.pk
Course Title	Theory of Automata	**Program**	BS Computer Science,
Course code	COSC-2105	**Credit Hours**	3

Course Outline: Finite State Models: Language definition preliminaries, formal languages and automata theory, regular expressions and regular languages, finite automata (FAs), transition graphs (TGs), NFAs, Kleene's theorem, formal modeling, language recognition, introductory logic and formal reasoning, transducers (automata with output), pumping lemma and non-regular languages.

Grammars and PDA: Context-free grammars, derivations, derivation trees and ambiguity, simplifying context-free languages (CFLs), normal-form grammars and parsing, algorithmic decision problems for context-free languages, closure properties and their implications for language recognition. Context-sensitive languages, grammars and linear bounded automata (LBA), Chomsky's hierarchy of grammars, and unrestricted (Type-0) grammars.

Decidability and Computability: Decidability and undecidability, computability of word functions, recursive enumerability, the Halting Problem, Rice's theorem, reductions, and the relationship between decidability and undecidability. Primitive-recursive and mu-recursive functions as formal models of computable functions.

Turing Machines Theory: Turing machines, Post machine, variations on Turing machines (TMs), TM encoding, Universal Turing Machine, defining computers by TMs, and Church–Turing thesis.

Week	Objective / learning outcomes	Content / sub content	Teaching methodology / activity	Assessment method / technique
1	Automata Theory & Formal Languages Introduction (Chap#1)	Computation, Formal Languages & its Types, What, Why, Where Automata Theory, Real Applications of Automata Theory, Basic constructs: Strings, Words, Alphabet, Language, Languages & Grammars, Basic Language Examples, Formal Modeling, Introductory Logic and Formal Reasoning	Activity based learning	Assignment, Quiz
2	Descriptive Definition of Languages (Chap#2) Recursive Definition of Languages (Chap#3)	Empty/Null String, Valid/Invalid Alphabets, KLEENE Star Closure & KLEENE Plus Closure, Defining Interesting Languages: EQUAL, EVEN-EVEN, INTEGER, factorial, FACTORIAL, DOUBLEFACTORIAL, SQUARE, DOUBLESQUARE, PRIME, PALINDROME etc., Basic Language Recognition Concepts	Activity based learning	Assignment, Quiz

3	Regular Expression (RE) Definition of Languages (Chap#4)	Recursive Definition of RE, RE Basic Constructs, Defining Interesting Languages by using Regular Expression, Language Operators (Union, Concatenation, Star), Operator Precedence, DFA to RE Construction, Algebraic Laws of RE, Some RE language Proofs	Activity based learning	Assignment, Quiz
4	Finite Automata (FA) Definition of Languages (Chap#5)	FA Definition & its basic construct (Transition table, Transition diagrams, start state, final state, transition function, dead state etc.), Defining Interesting Languages by using Finite Automata (starting with specific letters, ending with specific letter, letters occurring specific number of times etc.)	Activity based learning	Assignment, Quiz
Week	**Objective / learning outcomes**	**Content / sub content**	**Teaching methodology / activity**	**Assessment method / technique**
5	Acceptance, Rejection & Language Recognition cases in DFAs	Acceptance and rejection of strings, Language recognition using DFA, Designing DFAs for specified language conditions, Testing strings for membership, Formal modeling of simple language-recognition problems	Activity based learning	Assignment, Quiz
6	Transition Graph (TG) (Chap#6)	Defining Interesting Languages by using Transition Graph (TG), Generalized Transition Graph (GTG), Defining Interesting Languages by using Generalized Transition Graph (GTG), Basic Introduction to KLEENE's Theorem	Activity based learning	Assignment, Quiz
7	KLEENE's Theorem (Chap#7)	KLEENE's Theorem Part-I, KLEENE's Theorem Part-II, KLEENE's Theorem Part-III, Relationship between regular expressions, finite automata and regular languages	Activity based learning	Assignment, Quiz
8	Non-determinism (Chap#8)	Non-Deterministic Finite Automata (NFA or NDFA), NFA language recognition, Comparing deterministic and non-deterministic acceptance, NFA-based language modeling	Activity based learning	Assignment, Quiz
9	Finite Automata with Output (Chap#9)	Moore Machine, Mealy Machine, Machine Examples, Conversion b/w Machines, Conversion from Moore to Mealy, Conversion from Mealy to Moore	Activity based learning	Assignment, Quiz
10	Regular Languages (Chap#10)	Regular Operations (Union, Concatenation, Kleene Star), Elementary Questions about Regular Languages/Decision properties of Regular Languages, Properties of Regular Languages, Pumping Lemma (Prove a language is not regular), Closure properties of Regular Languages, Equivalence & Minimization of DFA's	Activity based learning	Assignment, Quiz
11	Non-Regular Languages (Chap#11) Context-Free Grammars (Chap#13)	Pumping Lemma, Myhill Nerode Theorem, CFG, Basic Components, Derivations, Ambiguity, Parse Tree etc., Language generation and recognition using CFGs Closure properties of context-free languages and algorithmic decision problems concerning context-free languages.	Activity based learning	Assignment, Quiz

12	Trees (Chap#14)	Basic Tree Structure, Polish Notation, Ambiguous CFG, Regular Grammar Basics, TG for Regular Grammar, Semi-word etc., Null, Nullable, Unit Productions, CNF, Derivation Trees and Parsing	Activity based learning	Assignment, Quiz
13	Push Down Automata (Chap#17)	Push Down Stack, PDA, Non-Deterministic PDA, PDA corresponding to CFG, CFG corresponding to PDA, PDA for Context-Free Language Recognition	Activity based learning	Assignment, Quiz
14	CFG=PDA, Context Free languages	PDA converted form, Equivalence of PDAs & CFG, Converting CFG to PDA & from PDA to CFG, Deterministic PDA (DPDA), PDA converted form, Equivalence of PDAs & CFG, Converting CFG to PDA & from PDA to CFG, Deterministic PDA (DPDA)	Activity based learning	Assignment, Quiz
Week	**Objective / learning outcomes**	**Content / sub content**	**Teaching methodology / activity**	**Assessment method / technique**
15	(Non-CFL, Parsing, Decidability & Undecidability) Context-Sensitive Languages, Grammars and LBA	Non-Context Free Languages (Chap#20), Non-CFL, Live production, dead production, Theorem, self-embedded non-terminal, Pumping Lemma for CFLs, Context-sensitive languages and grammars, Linear Bounded Automata (LBA), Chomsky's hierarchy of grammars, Decidability, Undecidability and introductory undecidability concepts. Recursive enumerability, computability of word functions, the Halting Problem, Rice's theorem, reductions, and unrestricted (Type-0) grammars.	Activity based learning	Assignment, Quiz
16	Turing Machines Theory	Turing machines with all its Components, TM language recognition, Post machine, Variations on TM, TM encoding, Universal Turing Machine, Defining Computers by TMs, Church-Turing thesis and the Church-Turing view of computation. Primitive-recursive and mu-recursive functions and their relationship to formal computability.	Activity based learning	Assignment, Quiz

Recommended Books:

- Daniel I. A. Cohen, "**Introduction to Computer Theory**", Second Edition.
- Sikander H. Khiyal, "***Theory of Automata and Computation***", National Book Foundation, 2004
- Nasir S.F.B and P.K. Srimani, "***A Textbook on Automata Theory***", Cambridge University Press, India, 2008

Additional Reading:

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Instructor's Signature

Khwaja Fareed University of Engineering & IT
Rahim Yar Khan

Tentative Course Plan
Department of Mechanical Engineering

Instructor	Ms. Arifa Arshad	**Email:**	arifa.arshad@kfueit.edu.pk
Course Title	Professional Ethics	**Program**	BS-COSC-8B
Course Number	SSCI-2111	**Credit Hours**	02
Lecture Time			

Course Introduction:

This course examines professional ethics, normative ethical frameworks, moral reasoning, and ethical decision-making in professional and organizational contexts. It also covers professional responsibilities, technology-related ethical issues, their social consequences, risk and liability, and professional practice.

COURSE LEARNING OUTCOMES (CLOs):

CLO 1: Demonstrate knowledge and understanding of prominent normative ethics frameworks, ethical principles, methods of moral reasoning, and their importance in professional conduct.

CLO 2: Apply ethical concepts and principles to ethical problems in professional, organizational, business, Computer Science, and Information Technology contexts, including issues related to the development and deployment of technological systems.

CLO 3: Analyze ethical challenges and dilemmas using normative ethics frameworks and ethical reasoning, including the ethical evaluation of technology, its social consequences, professional risks and liabilities, and introductory considerations of risk and uncertainty associated with emerging technologies.

CLO	Domain	Taxonomy Level	PLO
CLO 1	Cognitive	2	PLO 1
CLO 2	Cognitive	3	PLO 12
CLO 3	Cognitive	4	PLO 8

Course Content:

1. **Foundations of Professional Ethics:** Purpose of professional ethics, normative frameworks, ethical principles, moral reasoning, and professional conduct.
2. **Professional Responsibility and Standards:** Professional responsibility, liability, design standards, trust, codes of ethics, and compliance.
3. **Ethical Problem Analysis and Decision-Making:** Problem framing, utilitarian thinking, respect for persons, ethical analysis, case studies, and resolving dilemmas.
4. **Organizational and Business Ethics:** Organizational influence, business ethics, corporate culture, and ethical decision-making in organizations.
5. **Technology, Society, and Values:** Social and value dimensions of technology, opportunities and conflicts, ethical issues in CS/IT development and deployment, and their social consequences.
6. **Ethics in Technology Development and Deployment:** Ethical considerations in the design, implementation, and deployment of IT systems and their effects on society.
7. **Ethical Compliance and Professional Conduct:** Trust, reliability, ethical codes, compliance programs, delegation, communication, and monitoring of standards.
8. **Risk, Liability, and Emerging Technologies:** Professional risk and liability, public and regulatory approaches, uncertainty in emerging technologies, and possible future risks.
9. **Engineering and Management Decisions:** Ethical engineering and management decisions and professional responsibilities.
10. **Professional, Environmental, and Global Ethics:** Professional and environmental ethics, global ethical issues, business culture, and global ethical frameworks.
11. **International Literature and Computer Science Applications:** Relevant international literature and Computer Science applications related to professional and technology-related issues.

Text Book	*Engineering Ethics: Concepts and Cases* by C. Harris, Wordsworth, 1994.
Reference Books:	• Ferrell, O. C., and Fraedrich, John, *Ethical Decision Making and Cases*, New York: Houghton Mifflin. • Joseph W. Weiss, *Business Ethics: A Stakeholder & Issues Management Approach*, latest edition. • Lawrence M. Prentice, *Contemporary Moral Issues*.

SEQUENCE OF TOPICS TO BE COVERED

Lecture Date	Session #	16-WEEK LECTURE BREAKDOWN	Chapter #	
Week 1	**1**	Introduction to Professional Ethics, why study Professional Ethics? Role and purpose of professional ethics, frameworks for studying Professional Ethics.		
Week 2	**2**	Prominent normative ethics frameworks, ethical principles, moral reasoning, and their importance in professional conduct.		
Week 3	**3**	Framing ethical problems, clarifying concepts, utilitarian thinking, respect for persons, ethical analysis and case-based reasoning.		
Week 4	**4**	Responsibility in the profession, professional liability, design standards, trust and reliability, codes of ethics and compliance standards.		
Week 5	**5**	Resolving ethical problems, ethical decision-making, how organizations influence ethical decisions, organizational structure and business ethics.		
Week 6	**6**	Corporate culture in ethical decision-making, implications of organizational relationships for ethical decisions, management responsibility for ethical compliance programs.		
Week 7	**7**	The social and value dimensions of technology, the role of opportunity and conflict of opportunity, ethical issues in Computer Science and Information Technology development and deployment.		
Week 8	**8**	Revision + Quiz		
MID Term Exam				
Week 9	**9**	Ethical evaluation of technology, social and value consequences of technological systems for individuals, professional practice, and society.		
Week 10	**10**	Risk and liability in the profession, public approaches to risk, government-regulator approaches to risk, professional responsibility in technology-related activities.		
Week 11	**11**	Introductory discussion of risk and uncertainty in emerging technologies and Computer Science developments, possible consequences of technological change.		

Week 12	**12**	Introductory consideration of future risks arising from technological developments, professional responsibility when dealing with uncertain technological consequences.		
Week 13	**13**	Trust and reliability, codes of ethics and compliance standards, delegation of authority, effective communication of ethical standards, systems for monitoring ethical compliance, proper engineering and management decisions.		
Week 14	**14**	The professional and the environment, ethical responsibilities in professional practice, ethical issues around the globe, culture as a factor in business, adapting ethical systems to a global framework.		
Week 15	**15**	Engagement with relevant international literature and Computer Science application contexts, professional and technology-related ethical considerations.		
Week 16	**16**	Review of Professional Ethics frameworks, professional responsibilities, organizational ethics, technology-related ethical issues, risk and liability, case-based discussion. + Quiz + Revision		
Final Term Exam				

Student Evaluation criteria:

Attendance	5%
Workshop / Assignments/Case study	5%
Surprise Test/Sudden Test, Quizzes	5%
Class Participation	5%
Mid Term Paper	25%
Final Term paper	50%
Total	**100%**

Student Responsibilities:

Students must attend class. Failure to attend class may result in failure in the course. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments, students are responsible for knowing what they say. Students are also responsible for doing all assigned work on time. Excessive absences (more than 04) will result in “F Grade”. Students may prepare Sketchbook for taking notes and for references.

Instructor/Tutor

Approved by:

Chairman/ HOD/ Subject Specialist

KHWAJA FAREED UEIT RAHIM YAR KHAN | Institute of Computer Science

Institute of Computer Science

Course Outline

COSC-3104 Compiler Construction (3 Credit Hours)

Pre-Requisite: Theory of Automata
Program: BS Computer Science
Total Lectures: 32
Total Tutorials: 0
Instructor: Mr. Safdar Hussain
Email: safdar.hussain@kfueit.edu.pk

Course Introduction

This course provides a theoretical and systematic foundation for compiler construction as a formal process of translating programs between languages. Students study compiler design through Formal Languages, regular expressions and finite automata for lexical analysis; context-free grammars and parsing theory for syntax analysis; formal semantics, attribute grammars and type systems for semantic analysis; intermediate representations and syntax-directed translation for intermediate code generation; and program transformation, code optimization and target code generation. The course also develops error handling and recovery techniques, symbol-table management, run-time organization, and formal reasoning about syntax, semantics and semantics-preserving translation.

Course Contents

1. Compiler, interpreter, translator, phases of a compiler, front end and backend, analysis-synthesis model, source/target languages, passes of a compiler, bootstrapping, compiler organization, abstract machines, and interpretation models as foundations for understanding program execution and the runtime behavior of compiled and interpreted languages.
2. Alphabets, strings, languages, grammars, formal language definitions, regular languages, context-free languages, Chomsky hierarchy, BNF/EBNF, lexical and syntactic specifications, language recognition, and formal modeling of programming languages.
3. Tokens, lexemes and patterns, lexical rules, symbol classes, regular definitions, scanner design, lexical errors, token recognition, buffering, interface with syntax analysis and symbol tables, and functional approaches to scanner construction alongside regular-expression and finite-automaton-based methods.
4. Regular expressions, regular languages, DFA and NFA, transition diagrams, epsilon-NFA, regular expression to finite automata, NFA to DFA conversion, DFA minimization, and use of finite automata in lexical analyzer construction.
5. Context-free grammars, derivations, parse trees, ambiguity, associativity, precedence, left recursion, left factoring, grammar transformation, and syntax error detection.
6. Top-Down Parsing, recursive-descent parsing, backtracking, FIRST and FOLLOW sets, Predictive Parsing, LL(1) grammars, parsing-table construction, and functional approaches to parser construction.
7. Shift-reduce parsing, handles and viable prefixes, LR parsing, LR(0) items, SLR(1), CLR/LR(1), LALR(1), parsing tables, shift-reduce/reduce-reduce conflicts, and parser generators such as YACC/Bison.
8. Static semantics, syntax-directed definitions and translation, attribute grammars, inherited and synthesized attributes, symbol tables, scopes and bindings, type systems, type expressions, structural and name equivalence, type checking, coercion, semantic consistency, and fixed-point concepts and systems of equations related to programming-language semantics and analysis.
9. Abstract syntax trees, DAGs, three-address code, quadruples, triples, control-flow graphs, basic blocks, syntax-directed translation, translation of assignments, Boolean expressions, conditional and loop statements, procedure calls, backpatching, and transitive closures in program analysis and transformation.
10. Activation records, run-time stack, static and dynamic storage, parameter passing, procedure/function calls, non-local references, heap organization, run-time support, runtime memory management, garbage collection, and its relationship to storage allocation and heap management.
11. Optimization goals and constraints, local and global optimization, constant folding, algebraic simplification, common subexpression elimination, copy propagation, dead-code elimination, code motion, strength reduction, loop optimization, data-flow analysis, fixed-point computation and systems of equations in program analysis, basic-block optimization, and peephole optimization, with emphasis on semantics preservation and program transformation.
12. Target-machine model, instruction selection, addressing modes, register allocation and assignment, code generation from intermediate code and DAGs, control-flow translation, instruction ordering, and simple machine-dependent optimization.
13. Lexical, syntax and semantic error detection; panic-mode and phrase-level recovery; error reporting; recovery strategies; formal reasoning about language translation; semantic preservation; and correctness considerations in compiler transformations.

CLO	Course Learning Outcomes	Bloom's Taxonomy
CLO_1	Explain the theoretical foundations of compiler design using formal languages, regular expressions, finite automata, grammars, parsing theory and formal language specifications.	C2 (Understanding)
CLO_2	Analyze and construct lexical and syntax analyzers using regular expressions, finite automata, context-free grammars, Top-Down Parsing and Predictive Parsing techniques.	C4 (Analyzing)
CLO_3	Design semantic analyses, type-checking systems, intermediate representations, intermediate code generation, code optimization and code generation while applying formal reasoning and semantics-preserving translation.	C6 (Creating)

Assessment Items	Percentage
Quizzes	5%
Assignments	5%
Compiler Implementation Project	15%
Midterm Exam	25%
Final Exam	50%

Text and Reference Books

- A. V. Aho, M. S. Lam, R. Sethi and J. D. Ullman, Compilers: Principles, Techniques, and Tools, 2nd Edition, Addison-Wesley.
- Alfred V. Aho and Jeffrey D. Ullman, Principles of Compiler Design, Addison-Wesley.
- Cooper and Torczon, Engineering a Compiler, 2nd Edition, Morgan Kaufmann.
- Andrew W. Appel, Modern Compiler Implementation in C, Cambridge University Press.

Computer / Tool Usage

Students may use lexical and parser generator tools such as Lex/Flex and YACC/Bison, along with C/C++ or Python for assignments and the semester compiler project.

Week	Topics
Week 1	• Compiler design overview: compiler, interpreter and translator • Phases of compilation, front end and back end, analysis-synthesis model • Formal translation of source programs and introduction to compiler passes
Week 2	• Formal Languages: alphabets, strings, languages and grammars • Regular languages, context-free languages and Chomsky hierarchy • Formal specification of programming-language syntax and language recognition
Week 3	• Lexical analysis: tokens, lexemes, patterns and regular definitions • Regular expressions for token specification • Lexical errors, scanner design and symbol-table interface • Functional scanner construction.
Week 4	• Finite automata for lexical analysis: DFA and NFA • Transition diagrams and epsilon-NFA • RE to FA, NFA to DFA conversion and DFA minimization
Week 5	• Syntax analysis and context-free grammars (CFGs) • Derivations, parse trees and ambiguity • Associativity, precedence, left recursion and left factoring
Week 6	• Top-Down Parsing and recursive-descent parsing • FIRST and FOLLOW sets • Predictive Parsing and LL(1) parsing-table construction • Functional parser construction.
Week 7	• Bottom-up parsing and shift-reduce method • LR(0), SLR(1), LR(1)/CLR and LALR(1) parser theory • Parsing conflicts, parser construction and YACC/Bison overview
Week 8	• Semantic analysis: static semantics and syntax-directed definitions • Attribute grammars, inherited and synthesized attributes • Symbol tables, scopes, bindings and semantic checks
Week 9	**Mid Term Exam**
Week 10	• Type checking and type systems • Type expressions, structural and name equivalence, type conversion/coercion • Overloading, scope checking and semantic consistency
Week 11	• Intermediate representations: AST, DAG, three-address code • Quadruples, triples, basic blocks and control-flow graphs • Intermediate code generation for assignments and expressions
Week 12	• Syntax-directed translation for Boolean expressions and control flow • Backpatching for conditional and loop statements • Intermediate code generation for procedures and function calls
Week 13	• Run-time organization: activation records, stacks and heaps • Parameter passing and procedure calls • Storage allocation and run-time support for translated programs • Garbage collection, heap objects, and storage management.
Week 14	• Code optimization: goals, constraints and semantics preservation • Local and global optimizations: constant folding, common subexpressions, copy propagation and dead-code elimination • Loop optimizations, data-flow analysis and basic-block optimization
Week 15	• Code generation and target-machine model • Instruction selection, addressing modes and register allocation • Code generation from intermediate code/DAGs and peephole optimization
Week 16	• Error handling and recovery: lexical, syntax and semantic errors • Panic-mode and phrase-level recovery, error reporting and recovery strategies • Formal reasoning about translation and compiler correctness
Week 17	• Semester Project Presentations and Final Review • Integrated review: lexical analysis to code generation • Discussion of semantics-preserving translation, optimization and implementation issues • Abstract machines and interpretation models. • Fixed points, systems of equations, and transitive closures in program analysis and transformation.
Week 18	**Final Exam**

KHWAJA FAREED UEIT RAHIM YAR KHAN | Faculty of Information Technology & Management Sciences

Tentative Weekly Course Plan

Course Title	Linear Algebra		
Course Code	MATH-2103	**Credit Hour**	3
Name	Mr. Ubaid-ur-Rehman Asghar	**Email**	*ubaid.rehman@kfueit.edu.pk*

Course Contents (Core Topics):

- Systems of Linear Equations and Matrices: Gaussian Elimination, Matrix Inverses, Determinants (with proof of properties).
- Vector Spaces and Subspaces: Formal definition of Vector Spaces, Subspaces, Linear Span, Linear Independence and Dependence (with proofs of theorems).
- Basis and Dimension: Formal proof of Existence of Basis, Change of Basis, Rank and Nullity Theorem.
- Abstract Algebraic Foundations: Groups, fields, number systems, and field properties relevant to vector spaces and linear algebra.
- Linear Transformations: Formal definition and properties of Linear Transformations, Kernel and Range (with proofs).
- Eigenvalues and Eigenvectors: Calculation and theoretical properties of Eigenvalues and Eigenvectors, Diagonalization.
- Inner Product Spaces: Orthogonalization (Gram-Schmidt process) and Projection.
- Positive-Definite Matrices and Linear Systems: Positive-definite matrices, linear systems, and LU decomposition linking linear algebra with numerical linear algebra.

Resource Material	**1. Reference Books**	
	i	Linear Algebra and Its Applications by *David C. Lay* or *Gilbert Strang*
	ii	Elementary Linear Algebra (Applications Version) by Howard Anton
	2. Web Resources	
	i.	https://3blue1brown.com/topics/linear-algebra
	ii.	https://khanacademy.org/math/linear-algebra

TOPICS TO BE COVERED

Week #	Topics (outline of main topics and sub topics)	Tutorial /Laboratory
1	Real and complex numbers, Introduction to Matrices and Systems of Equations, Number systems, groups, fields, and field properties of vector spaces	Tutorial
2	Gaussian Elimination, Gauss-Jordan Elimination, Matrix Operations, LU decomposition for solving linear systems	Tutorial
3	Determinants and their Theoretical Properties (Proof of basic theorems)	Tutorial
4	Formal Definition of Vector Spaces and their Subspaces	Tutorial

5	Linear Combination, Span, and Proof of Linear Independence/Dependence	Tutorial
6	Basis and Dimension of a Vector Space	Tutorial
7	Row Space, Column Space, and Null Space	Tutorial
8	Linear Transformations and Linear Mappings	Tutorial
9	Definition, Matrix Representation, Properties (Kernel and Range)	
Mid Term Exam		
10	Inner Product Spaces, Orthogonality, Orthogonal Complements	Tutorial
11	Gram-Schmidt Orthogonalization Process	Tutorial
12	Eigenvalues and Eigenvectors (Calculation and Geometric Interpretation)	Tutorial
13	Diagonalization and its applications	Tutorial
14	Mathematical reasoning and Applications of Linear Algebra relevant to Computer Science	Tutorial
15	Review and Introduction to Symmetric Matrices	Tutorial
16	Positive-definite matrices and systems, quadratic forms, and numerical linear algebra.	Tutorial
Final Term Exam		

Student Evaluation criteria:

Assignments	5%
Surprise Test, Quizzes	5%
Presentation	10%
Mid Term Paper	30%
Final Term paper (***Marks Distribution:*** 20 % from Mid Term and 80% from Final Term), i.e. 10 + 40=50	50%
Total	**100%**

Student Responsibilities:

A student must have attended at least **75%** of the classes held in a course in order to be allowed to sit in the final examination. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments. Students are also responsible for doing all assigned work on time. Students may prepare Sketchbook for taking notes and for references.

Course Outline

MATH-1100 CALCULUS (3 Credit Hours)

Pre-Requisite:
Course Title: Calculus
Code: MATH-1100
Credit Hours: 3
Instructor: Mr. Nasir Nadeem
Office: Mechanical Engineering Department, KFUEIT
Email: *nasir.nadeem@kfueit.edu.pk*
Office Hours: 10:30am - 12:30 pm

Course Objective

This foundational course provides rigorous study in Mathematical Analysis, focusing on theoretical concepts, formal definitions, and proof techniques essential for higher-level quantitative studies, including an introduction to multivariate functions

Course Contents

Foundations of Analysis: Limits, continuity, sequences, and series with emphasis on ε - δ proofs (to satisfy the 'proofs' requirement).
Elementary and Hyperbolic Functions: Hyperbolic and inverse hyperbolic functions, their basic derivatives and integrals.
Differential Calculus: Functions and Inverse Functions, Techniques of differentiation, Mean Value Theorem, Taylor/Maclaurin Series.
Improper Integrals and Convergence: Improper integrals, convergence, and evaluation techniques.
Vector and Multivariate Calculus: Functions of several variables, Partial Derivatives (First and Second Order), Multiple Integrals (Double and Triple), and basic concepts of Vector Fields.
Multivariable Analysis: Coordinate systems in (\mathbb{R}^n), vector-valued mappings, sequences and limits, directional derivatives, and total differentials.
Multivariable Taylor Expansion and Newton Methods: Taylor expansion and Newton-type methods for multivariable problems.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
1	Students will be able to apply formal mathematical proof techniques to demonstrate the properties of limits and continuity	-	-
2	Students will be able to differentiate and integrate functions of several variables, demonstrating proficiency in Partial Derivatives and Multiple Integration	-	-
3	Students will analyze complex problems using Taylor/Maclaurin Series and apply fundamental theorems of calculus to various domains	-	-

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes / Assignments	-	-	-
Mid semester Exam	-	-	-
Final Exam	-	-	-

Overall Grading Policy

Assessment Items	Percentage
Sessional (Quizzes + Assignments + ppts)	20%
Midterm Exam	30%

Final Exam	50%

Text and Reference Books

Text books:

- Calculus: Early Transcendentals by *James Stewart*
- Thomas' Calculus by *George B. Thomas, Jr.* et al

Reference books:

- Introduction to Real Analysis by *Robert G. Bartle*

Administrative Instructions

- According to institute policy, 75% attendance is *mandatory* to appear in the final examination.
- Assignments and report must be submitted as per announced format and within due date.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Lecture Breakdown

Week 01	Introduction to Real Numbers, Functions, Inverse Functions, Definition of Limits ($\varepsilon \rightarrow \delta$)
Week 02	Continuity, Inverse Functions, Differential Calculus, Basic Rules of Differentiation
Week 03	Chain Rule, Implicit Differentiation, Related Rates
Week 04	Theorems of Differentiation (Rolle's Theorem, Mean Value Theorem, L'Hôpital's Rule), Hyperbolic functions, Newton and secant methods for root-finding.
Week 05	Integration and Integral Calculus, Definite and Indefinite Integrals, Fundamental Theorem of Calculus.
Week 06	Techniques of Integration (Substitution, By Parts), Improper integrals and their convergence.
Week 07	Applications of Integrals (Area, Volume)
Week 08	Sequences and Infinite Series, Convergence and Divergence Tests, Taylor Expansion
Week 09	**Mid Term Exam**
Week 10	Introduction to Multivariate Functions
Week 11	Level Curves and Limits
Week 12	Continuity in Higher Dimensions, Coordinate systems in (\mathbb{R}^n), vector-valued mappings, sequences, limits, directional derivatives, and total differentials.
Week 13	Partial Derivatives (First Order), Chain Rule for Multiple Variables
Week 14	Second Order Partial Derivatives, Mixed Partial Derivatives, Critical Points, Multivariable Taylor expansion and Newton-type methods.
Week 15	Multiple Integrals (Double Integrals over Rectangular and General Regions)
Week 16	Change of Variables in Multiple Integrals (Polar, Cylindrical coordinates)
Week 17	Introduction to Vector Fields, Applications of Calculus to Mathematical and Computational Problems
Week 18	**Final Term Exam**

KHWAJA FAREED UEIT RAHIM YAR KHAN | Electrical Engineering

Probability and Statistics

Credit Hours:	3	Course Code	STAT-2104	Instructor	Mr. Muhammad Hashir Zubair

Objectives:

This course provides a mathematical and analytical foundation in probability and statistics, with emphasis on their theoretical principles and practical applications. It develops rigorous understanding of uncertainty, random phenomena, statistical modeling, estimation, inference, and regression, with emphasis on concepts used in data analysis.

Course Learning Outcomes:

- Students will understand fundamental principles of probability theory, statistical reasoning, and uncertainty modeling.
- Students will formulate and analyze random variables, probability distributions, expectations, variances, and joint distributions.
- Students will apply Bayes' theorem, conditional probability, and independence to probabilistic reasoning and data-driven problems.
- Students will perform descriptive and inferential statistical analysis, including estimation, confidence intervals, and hypothesis testing.
- Students will develop and interpret regression models and apply statistical methods to problems in CS.

Course Contents:

- Introduction to Probability and Statistics: populations and samples, variables, data types, measurement scales, statistical reasoning, and applications in Computer Science.
- Descriptive Statistics: measures of central tendency, dispersion, quartiles, percentiles, skewness, kurtosis, frequency distributions, and graphical summaries.
- Probability Theory: sample spaces, events, set operations, counting principles, permutations, combinations, axioms of probability, and basic probability rules.
- Conditional Probability, Independence, and Bayes' Theorem: conditional probability, multiplication rule, law of total probability, independence, Bayes' theorem, and Bayesian reasoning.
- Random Variables: discrete and continuous random variables, probability mass functions, probability density functions, cumulative distribution functions, and transformations.
- Probability Distributions: Bernoulli, Binomial, Geometric, Negative Binomial, Hypergeometric, Poisson, Uniform,

Exponential, Gamma, Normal, and related distributions.

- Expectation and Variance: expected value, moments, variance, standard deviation, linearity of expectation, conditional expectation, variance rules, and moment-generating concepts.
- Joint Random Variables: joint, marginal, and conditional distributions, covariance, correlation, independence, conditional expectation, and functions of random variables.
- Random Vectors and Independent Sums: Random vectors, transformations, general characteristics, and sums of independent random variables.
- Sampling and Limit Results: sampling distributions, Law of Large Numbers, Central Limit Theorem, normal approximation, and statistical foundations for large datasets.
- Statistical Estimation: point estimation, estimators, bias, consistency, efficiency, maximum likelihood estimation, least-squares estimation, and sampling distributions of estimators.
- Multivariate Probability Distributions: Multivariate normal distributions and characteristic functions.
- Interval Estimation: confidence intervals for means, proportions, variances, and regression parameters; interpretation and uncertainty quantification.
- Statistical Analysis and Inference: null and alternative hypotheses, test statistics, p-values, significance levels, Type I and Type II errors, statistical power, z-tests, t-tests, chi-square tests, and basic ANOVA.
- Regression Analysis: simple and multiple linear regression, least-squares fitting, model parameters, residual analysis, coefficient interpretation, prediction, confidence intervals, and model assumptions.
- Statistical Modeling : correlation analysis, model diagnostics, uncertainty in predictions, feature relationships, statistical decision-making, and interpretation of empirical results.
- Advanced Probability Limits: Bernoulli scheme and Poisson limit theorem.
- Measure-Theoretic Foundations: Sigma-algebras, measures, measurability, and integration in probability theory.

Text Books:

Sheldon M. Ross, A First Course in Probability
Ronald E. Walpole et al., Probability and Statistics for Engineers and Scientists

Recommended Books:

George Casella and Roger L. Berger, Statistical Inference
Douglas C. Montgomery and George C. Runger, Applied Statistics and Probability for Engineers

Weekly Plan

Sr. No	Weekly	Topics
1	Week 1st	Introduction to Probability and Statistics; populations, samples, variables, data types, descriptive statistics, measures of central tendency and dispersion
2	Week 2nd	Probability theory; sample spaces, events, set operations, counting principles, permutations, combinations, axioms and basic probability rules
3	Week 3rd	Conditional probability, independence, law of total probability, and Bayes' theorem; probabilistic reasoning and applications
4	Week 4th	Random variables; discrete and continuous random variables, PMF, PDF, CDF, transformations, and probability calculations
5	Week 5th	Probability distributions I; Bernoulli, Binomial, Geometric, Negative Binomial, Hypergeometric, and Poisson distributions, Bernoulli scheme and Poisson limit theorem.
6	Week 6th	Probability distributions II; Uniform, Exponential, Gamma, Normal distributions; standardization and distributional relationships
7	Week 7th	Expectation and variance; moments, linearity of expectation, conditional expectation, covariance, correlation, and joint random variables, Independent random-variable sums, random vectors, transformations, and multivariate normal distribution.
8	Week 8th	Sampling distributions; Law of Large Numbers, Central Limit Theorem, normal approximation, and mid-course revision, Characteristic functions and their role in distributions and probability limits.
9	Week 9th	**Mid Term Exam**
10	Week 10th	Statistical estimation; point estimators, bias, consistency, efficiency, maximum likelihood estimation, least-squares estimation
11	Week 11th	Interval estimation; confidence intervals for means, proportions, variances, and regression parameters; interpretation of uncertainty
12	Week 12th	Statistical analysis and inference; hypothesis testing, p-values, significance levels, Type I/II errors, power, z-tests and t-tests
13	Week 13th	Chi-square tests, goodness-of-fit, independence tests, one-way ANOVA, and comparison of statistical groups
14	Week 14th	Regression analysis I; simple linear regression, least squares, coefficient estimation, correlation, residuals, assumptions, and prediction
15	Week 15th	Regression analysis II and applications; multiple linear regression, model diagnostics, confidence intervals
16	Week 16th	Final revision, integrated problem solving
17	Week 17th	Statistical interpretation, evaluation, Introduction to sigma-algebras, measures, measurability, and integration.
18	Week 18th	**Final Exam**

KHWAJA FAREED UEIT RAHIM YAR KHAN | Electrical Engineering

Course Outline

MATH-2122 Differential Equations (3 Credit Hours)

Pre-Requisite: Calculus
Course Title: Differential Equations
Course Code: MATH-2122
Credit Hours: 3
Instructor: Ms. Saba Jamil
Office: 2.21DR, 1st Floor, Electrical Engineering Building, KFUEIT, RYK
Email: *saba.jamil@kfueit.edu.pk*
Office Hours: 13:30 - 14:30

Course Objective

This course extends the concepts of Calculus by introducing advanced analytical techniques to model and solve real-world problems. It covers higher-dimensional systems and theoretical foundations essential for understanding complex mathematical models

Course Contents

First Order ODEs: Analytical solutions (Separable, Exact, Linear).

Higher Order ODEs: Homogeneous and non-homogeneous equations.

Numerical Initial-Value Problem Methods: Explicit and implicit Runge - Kutta methods, one-step and multistep schemes.

Vector and Higher-Dimensional Systems: Systems of Linear Differential Equations, Matrix methods for solving systems proofs of existence and uniqueness of solutions are a key focus).

Numerical Partial Differential Equations: Finite-difference, finite-element, and finite-volume methods.

Partial Differential Equations (PDEs): Introduction to first-order PDEs, boundary value problems, and Fourier Series.

Iterative Linear-System Solvers: Iterative methods for solving linear systems arising from discretized differential-equation models.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
1	Students will be able to apply advanced analytical methods to solve ordinary and systems of differential equations	-	-
2	Students will develop the ability to theoretically analyze the behavior of solutions, including stability and uniqueness (demonstrating analysis with proofs)	-	-
3	Students will be able to formulate simple real-world models using differential equations, including an introduction to higher-dimensional modeling	-	-

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes / Assignments	-	-	-
Mid semester Exam	-	-	-
Final Exam	-	-	-

Overall Grading Policy

Assessment Items	Percentage
Sessional (Quizzes + Assignments + ppts)	20%
Midterm Exam	30%
Final Exam	50%

Text and Reference Books

- **Text book:** Elementary Differential Equations and Boundary Value Problems by *Boyce and DiPrima*
- **Reference book:** Differential Equations by *Zill*

Administrative Instructions

- According to institute policy, 75% attendance is *mandatory* to appear in the final examination.
- Assignments and report must be submitted as per announced format and within due date.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Lecture Breakdown

Week 01	First-order ordinary differential equations (ODEs), Classification, Separable equations, solution methods, and First-Order Linear Equations (Integrating Factors)
Week 02	Solutions to First Order Linear Equations (Integrating Factors)
Week 03	Exact Equations and Numerical Methods for First Order ODEs
Week 04	Second-order linear differential equations, Homogeneous and non-homogeneous equations
Week 05	Method of Undetermined Coefficients and Variation of Parameters
Week 06	Series Solutions for ODEs (Taylor and Power Series)
Week 07	Laplace transforms and their application to solving differential equations
Week 08	Systems of Linear Differential Equations, Matrix Formulation
Week 09	Eigenvalues and Eigenvectors for systems of ODEs
Mid Term	
Week 10	Phase Plane Analysis and Stability of systems
Week 11	Introduction to Partial Differential Equations (PDEs), Classification
Week 12	Solutions of simple PDEs (Separation of Variables), Finite-difference methods for numerical PDEs.
Week 13	Fourier Series and Boundary Value Problems, Finite-element and finite-volume methods for numerical PDEs and boundary-value problems.
Week 14	Applications of Differential Equations in Complex Modeling and Data Analysis
Week 15	Numerical Methods for Solving Differential Equations and Error Analysis, Explicit and implicit Runge - Kutta methods, one-step and multistep schemes, and iterative solvers for linear systems arising from numerical differential-equation models.
Week 16	Presentations

Final Term

KHWAJA FAREED UEIT RAHIM YAR KHAN | Faculty of Engineering | Department of Mechanical Engineering

Course Outline

MATH-2124 Numerical Analysis (3 Credit Hours)
Course Title: Numerical Analysis **Course Code:** MATH-2124 **Credit Hours:** 3 **Instructor:** Ms. Huba Rana **Office:** Mechanical Engineering Department, KFUEIT **Email:** *huba.rana@kfueit.edu.pk* **Office Hours:** 14:30am - 15:30 pm (Tuesday)

Course Objective

This course is to introduce students to the mathematical concepts and computational techniques required to find approximate solutions to complex mathematical problems that lack analytical (exact) solutions. The course emphasizes understanding the algorithms, error analysis, and stability of numerical methods

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy
1	Apply and analyze various numerical methods (e.g., Bisection, Newton-Raphson) to efficiently find the roots of non-linear equations.	PLO-1	C3 (Applying) / C4 (Analyzing)
2	Solve systems of linear equations using direct (e.g., Gaussian Elimination) and iterative (e.g., Jacobi, Gauss-Seidel) techniques and understand their computational efficiency	PLO-2	C3 (Applying)
3	Use techniques like Lagrange and Newton interpolation to approximate complex functions, and apply the concept of Least Squares Approximation to fit data (fundamental in Machine Learning).	PLO-2	C3 (Applying)
4	Implement numerical methods for integration (Quadrature) and differentiation, and apply methods for solving Ordinary Differential Equations (ODEs) like the Runge-Kutta method.	PLO-1	C3 (Applying)

CLO Assessment Mechanism

Assessment tools	CLO_1	CLO_2	CLO_3
Quizzes / Assignments	-	-	-
Mid semester Exam	-	-	-
Final Exam	-	-	-

Overall Grading Policy

Assessment Items	Percentage
Sessional (Quizzes + Assignments + ppts)	25%
Midterm Exam	25%
Final Exam	50%

Text and Reference Books

Recommended books:

- Numerical Analysis by *Richard L. Burden and J. Douglas Faires*
- Numerical Methods for Engineers by *Steven C. Chapra and Raymond P. Canale*
- Introduction to Numerical Analysis by *S. S. Sastry*

Administrative Instructions

- According to institute policy, 75% attendance is *mandatory* to appear in the final examination.
- Assignments and report must be submitted as per announced format and within due date.
- For queries, kindly follow the office hours in order to avoid any inconvenience.

Lecture Breakdown

Week 01-02	Mathematical preliminaries, Sources of Errors (Round-off, Truncation), Absolute and Relative Errors, Stability and Conditioning of problems
Week 03-04	Solutions of nonlinear equations (Bisection, Newton-Raphson, Fixed-Point Iteration, Secant Method), convergence rates, multivariable Newton's method for systems of nonlinear equations, and comparison of convergence order with scalar root-finding.
Week 05-06	Direct linear systems (Gauss Elimination, LU Decomposition), positive-definite matrices and linear systems, numerical solution techniques, and computational properties.
Week 07-08	Iterative linear systems (Jacobi Method, Gauss-Seidel Method, SOR, Convergence Criteria), positive-definite systems, and convergence and computational properties of iterative solvers.
Week 09	Lagrange Interpolating Polynomials, Divided Differences, Newton's Form of the Interpolating Polynomial
Week 10	Numerical Differentiation & Integration (Finite Differences, Trapezoidal Rule)
Week 11	Numerical Differentiation & Integration (Simpson's Rules, Composite Rules, Gaussian Quadrature)
Week 12	Polynomial Approximation, Linear Least Squares Regression, Non-linear Least Squares
Week 13	Numerical ODEs: Initial Value Problems, Euler's Method, and systems of ordinary differential equations.
Week 14	Runge-Kutta Methods (RK2, RK4), explicit and implicit Runge-Kutta schemes, and one-step and multistep methods for systems of ODEs.
Week 15	Optimality and sensitivity, gradient and conjugate-gradient methods, Newton and quasi-Newton methods, line-search and trust-region methods.
Week 16	KKT conditions and regularity, penalty and barrier methods, augmented-Lagrangian methods, Lagrange-Newton, SQP, and nonlinear interior-point methods.

KHWAJA FAREED UEIT RAHIM YAR KHAN | Institute of Computer Science

Discrete Structures					
Credit Hours:	3	**Course Code**	COSC-1103	**Instructor**	Mr. Mujeeb Ur Rehman

Objectives:

This course introduces the fundamental mathematical and abstract structures used in Computer Science and Data Science. It develops logical reasoning, proof techniques, discrete problem-solving skills, combinatorial methods, and the study of sets, relations, functions, number theory, graphs, Boolean algebra, coding theory, and cryptographic foundations essential for algorithmic analysis and theoretical computer science.

Course Learning Outcomes:

CLO 1: Students will be able to apply mathematical logic, set theory, relations, functions, and formal proof techniques, including direct proof, contradiction, and mathematical induction, to solve discrete problems.
CLO 2: Students will be able to analyze and solve discrete mathematical problems using combinatorics, recurrence relations, number theory, Boolean algebra, and related mathematical structures.
CLO 3: Students will be able to model and analyze computational problems using graph theory, trees, relations, and other discrete structures, including fundamental graph properties and algorithms.
CLO 4: Students will be able to apply concepts of coding theory and cryptographic foundations, along with discrete mathematical reasoning, to basic computer science and data-oriented problems.

Course Contents:

Sets, relations and functions; propositional and predicate logic, logical equivalence, propositional normal forms, resolution, and proof techniques including induction, contradiction and direct proof; counting principles, binomial coefficients, counting estimates, permutations, combinations, partitions, pigeonhole principle, inclusion-exclusion, recurrence relations, Latin squares, Ramsey's theorem, Möbius inversion, formal power series, special number sequences, and combinatorial games; sequences and summation; number theory including prime numbers, divisibility and modular arithmetic; Boolean algebra and logic gates; graph theory, connectivity, trees, graph traversal, paths, circuits, Eulerian and Hamiltonian graphs, graph coloring, chromatic number, matching, flows, Hall's and König's theorems, Menger's theorem, planar graphs, Euler's formula, Kuratowski's theorem, graph duality, cycle bases, interval and chordal graph classes, graph treewidth, graph-class recognition and optimization examples; discrete mathematical problem solving; coding theory and cryptographic foundations, including information measures, source coding, and error-correcting-code concepts.

Text Books:

1. Discrete Mathematics and Its Applications by *Kenneth H. Rosen*

Recommended Books:

1. Discrete Mathematics by *Richard Johnsonbaugh*

Weekly Plan of "Discrete Structures" by Mr. Mujeeb Ur Rehman

Sr. No	Weekly	Topics
1	Week 1st	Introduction to Propositional Logic and Compound Propositions,
2	Week 2nd	Conditional Statements, Logical Equivalence, Propositional normal forms and resolution.
3	Week 3rd	Methods of Proof (Direct Proof, Contraposition, Contradiction)
4	Week 4th	Mathematical Induction
5	Week 5th	Set Theory, Set Operations, Set Identities (with proofs)
6	Week 6th	Functions, Sequences, Summation Notation, and Introduction to Recurrence Relations
7	Week 7th	Relations, Equivalence Relations, Partial Orderings
8	Week 8th	Number Theory: Prime Numbers, Modular Arithmetic, and Discrete Mathematical Problem Solving
Mid Term Exam		
9	Week 9th	Combinatorial Reasoning: Principles of Counting, Pigeonhole Principle, Inclusion-Exclusion
10	Week 10th	Binomial coefficients, counting estimates, partitions, advanced combinatorics, Permutations, and Combinations
11	Week 11th	Introduction to Graph Theory, Terminology, Connectivity, matching, flows, graph theorems
12	Week 12th	Planar graphs, graph duality, cycle bases, Isomorphism, Trees
13	Week 13th	Interval, chordal and planar graph classes, treewidth, and graph-class recognition and optimization, Eulerian and Hamiltonian Paths, Graph Coloring, Shortest Path Algorithms
14	Week 14th	Boolean Algebra and Logic Gates
15	Week 15th	Coding Theory and Cryptographic Foundations (Applications of Discrete Mathematics in Security), Information measures, source coding, and error-correcting codes.
16	Week 16th	Presentation
Final Term Exam		

KHWAJA FAREED UEIT RAHIM YAR KHAN | Institute of Computer Science

Course description

Final Year Project - I and II (6 Credit hours)

Course Title: Final Year Project - I and II
Course Code: COSC-4301 and COSC-4302
Credit Hours: 3 + 3
Instructor: Mr. Mutiullah
Office: 3.2L, 3rd Floor, Institute of Computer Science, ICT Building, KFUEIT, RYK
Email: *mutiullah.jamil@kfueit.edu.pk*
Office Hours: 10:30 - 11:30

Course Objective

This course provides a structured two-phase framework for the development of a final year project. The first phase focuses on problem identification, literature review, requirements analysis, feasibility assessment, project planning, system design, and preparation of technical documentation. The second phase focuses on implementing the proposed solution through programming, frameworks, module development, system integration, testing, debugging, and performance improvement. Throughout both phases, students work under the guidance of their supervisor through regular progress meetings and conclude the project with formal documentation, presentation, and evaluation before the project panel.

Course Contents

Course Contents

- Identification and analysis of a relevant problem, project scope, objectives, and expected outcomes.
- Literature review and analysis of existing solutions, technologies, and methodologies.
- Requirements gathering, feasibility analysis, development methodology, project planning, and identification of relevant ethical, societal, and responsible-use considerations.
- Preparation of Software Requirements Specification (SRS), system architecture, database design, interface design, and other technical documentation.
- Development and implementation of the proposed system using appropriate programming languages, frameworks, tools, and technologies. For perception and computer-vision projects, relevant 2D image features, color information, and motion features.
- Implementation and integration of project modules and system components, including model training where applicable.
- Testing, debugging, security validation, performance improvement, and system refinement; evaluation of results, model/system limitations, relevant risks, and potential effects on people or decisions.
- Preparation of user manual, final technical documentation, project report, and presentation, documenting the chosen model/system approach, implementation or training process, evaluation methods, results, limitations, and justification of conclusions.
- Final project demonstration, Viva/Question-Answer Session, and evaluation by the project panel.

Mapping of CLOs and PLOs

Sr. No	Course Learning Outcomes	PLOs	Blooms Taxonomy

1	Students will be able to analyze a problem, define project requirements, objectives, scope, and appropriate development methodology.	-	-
2	Students will be able to design and document the proposed system, including its architecture, modules, database, and user interface.	-	-
3	Students will be able to implement, integrate, test, debug, and improve the proposed system using appropriate technologies and development frameworks.	-	-
4	Students will be able to prepare and present the complete project documentation, demonstrate the developed system, and defend the project professionally before the evaluation panel.	-	-

CLO Assessment Mechanism

Assessment Tools	CLO_1	CLO_2	CLO_3	CLO_4
Supervisor Evaluation (Meetings, Progress, Implementation)	✓	✓	✓	-
Final Presentation & Demonstration	✓	✓	✓	✓
Viva / Question-Answer Session	-	-	✓	✓

Overall Grading Policy

Assessment Items	Percentage
Supervisor Evaluation (Progress, Meetings, Implementation)	30%
Final Presentation & Viva (Evaluation by Panel)	70%

Administrative Instructions

- Students must maintain regular contact with their supervisor and submit progress reports on time.
- The project must be implemented practically with working functionality and proper source code.
- All deadlines for prototype submission, documentation, and final demo must be strictly followed.
- Attendance and participation in project meetings are mandatory as per institute policy.
- Students must be prepared for final viva, demonstration, and question-answer session.

Project Breakdown

Week	Topics
Week 01	Project orientation, idea identification, and problem definition
Week 02	Project scope, objectives, and expected outcomes
Week 03	Background study and literature review

Week 04	Analysis of existing systems and proposed solution
Week 05	Requirements gathering and functional requirements
Week 06	Non-functional requirements and project constraints, Ethical/social context, project risks, responsible use, and potential impacts.
Week 07	Feasibility analysis and development methodology
Week 08	Technology selection and project development planning, Method / technology suitability, limitations, and relevant image, color, or motion features for vision projects.
Week 09	System architecture and high-level system design
Week 10	Database design and module/component design
Week 11	User interface / prototype design and workflow definition
Week 12	Software Requirements Specification (SRS) preparation
Week 13	Project report and technical documentation preparation
Week 14	Documentation review, refinement, and supervisor feedback
Week 15	Final documentation and presentation preparation
Week 16	FYP-I Final Presentation, Viva, and Panel Evaluation
Week 17	FYP-II implementation planning and development environment setup
Week 18	Core functionality development - Part 1
Week 19	Core functionality development - Part 2
Week 20	Module development and implementation
Week 21	Database integration and backend/system integration
Week 22	User interface implementation and refinement
Week 23	Integration of all major modules
Week 24	Initial system testing and validation; evaluation protocols and test cases.
Week 25	Debugging, error handling, and issue resolution; documentation of defects and corrective actions.
Week 26	Additional feature development and system optimization; evaluation of functional and performance improvements.
Week 27	Security validation and performance enhancement; security testing, performance results, and remaining risks.
Week 28	Comprehensive system testing and refinement; evaluation results and identification of system limitations.
Week 29	User manual and final technical documentation, including model/system selection, implementation/training approach, evaluation methods, results, limitations, and ethical / social considerations.
Week 30	Final code cleanup, system testing, and project completion; verification of the final implementation.
Week 31	Final project rehearsal and presentation preparation; presentation of results, limitations, and relevant ethical/social impacts.
Week 32	Final Demonstration, Viva, Panel Evaluation, and Submission.

Khwaja Fareed University of Engineering & IT
Rahim Yar Khan

Tentative Course Plan
Department of Computer Science & Information Technology

Instructor	Ms. Rida Fatima	**Email:**	fatima.rida55@gmail.com
Course Title	Operating System	**Program**	BS-INFT 4A
Course Number	Cocs-2102	**Credit Hours**	03
Lecture Time			

Course Objective:

Course Outcomes:

Methods of Teaching:

The main teaching method used in this course will be Direct teaching. Apart from that, students will be given assignments to explore computers and their uses in today's environment.

Resource Material

1. Silberschatz, P. B. Galvin, and G. Gagne, "Operating Systems Concepts (Essentials)", 9th Edition, John Wiley & Sons, 2012.
2. W. Stallings, "Operating Systems: Internals and Design Principles", 8th ed, Pearson, 2015.

2.Reference Book		3.Research Papers	
I		i	
Ii		ii	
4.Hot Research Papers		5.Web Resources	
I		i	
Ii		ii	

Office Help Hours: 8 to 4

SEQUENCE OF TOPICS TO BE COVERED

Lecture Date	Session #	Topics (outline of main topics and sub topics)	Chapter #	
Week 1	**1**	Basics of OS, Functions of OS, Computer system organization, Computer Operation and structure, Interrupts, I/O Structure, Direct memory Access structure, Storage structure, Operating system Architecture, Kernel Data Structure, process management system, Memory management system, Storage management system		
Week 2	**2**	Operating System Services, Operating System Interface user, System Calls, Types of System Calls, System Programs, Operating System Design and Implementation, Operating System Structure, Virtual Machines		
Week 3	**3**	**Process** Concept, Process Scheduling, Operations on Processes, Cooperating Processes, Interprocess Communication, Basics of **Threads**, Multithreading Models, Threading Issues, Pthreads		
Week 4	**4**	Inter process communication, Cooperating Processes, communication models, IPC Data Transfer Message Passing, Direct Communication, Indirect Communication, Synchronization, Sockets, Remote Procedure Calls		
Week 5	**5**	Basic Concepts **CPU scheduling**, Scheduling Criteria, Scheduling Algorithms, Thread Scheduling, Multiple-Processor Scheduling, Real-Time Scheduling		
Week 6	**6**	Scheduling Algorithms, First-Come, First-Served (FCFS) Scheduling, Shortest-Job-First (SJF) Scheduling, Priority Scheduling, Round Robin (RR), Algorithm Evaluation		

Week 7	**7**	**Process Synchronization** Background, The Critical-Section Problem, Peterson's Solution, Synchronization Hardware, Semaphores,		
Week 8	**8**	Revision + Quiz		
MID Term Exam				
Week 9	**9**	The **Deadlock** Problem, Deadlock Characterization, **Resource-Allocation Graph,** Methods for Handling Deadlocks, Deadlock Prevention		
Week 10	**10**	Deadlock Avoidance, Deadlock Detection, Bankers Algorithm, Recovery from Deadlock + quiz		
Week 11	**11**	**Memory Management** Background, Swapping, Contiguous Allocation, Paging, Segmentation, Segmentation with Paging, Demand Paging, Copy-on-Write, Page Replacement, Allocation of Frames, Thrashing.		
Week 12	**12**	**Virtual Memory,** Background, Demand Paging, Copy-on-Write, Page Replacement, Allocation of Frames, Thrashing, Memory-Mapped Files, Allocating Kernel Memory, Other Considerations, Operating-System Examples		
Week 13	**13**	**File management Concept**, Access Methods, Directory Structure, File-System Mounting, File Sharing, File-System Structure, File-System Implementation, Directory Implementation, Allocation Methods, Free-Space Management, Efficiency and Performance		
Week 14	**14**	Overview of **Mass Storage** Structure, Disk Structure, Disk Attachment, Disk Scheduling, Disk Management, Swap-Space Management, RAID Structure, Disk Attachment + Quiz		
Week 15	**15**	**Distributed system,** Types of Network based Operating Systems, Network Structure, Network Topology, Communication Structure, Robustness, Design Issues		
Week 16	**16**	**Protection and Security**, Policy and Mechanism of security, The Security Problem, Authentication Mechanisms, User Authentication, Network Authentication, Program Threats, System Threats, Protection System. + Quiz + Revision		
Final Term Exam				

Student Evaluation criteria:

Attendance	5%
Workshop / Assignments/Case study	5%
Surprise Test/Sudden Test, Quizzes	5%
Class Participation	5%
Mid Term Paper	30%
Final Term paper	50%
Total	**100%**

Student Responsibilities:

Students must attend class. Failure to attend class may result in failure in the course. Students must also arrive on time and remain in class for the entire period. Cellular Phones and Beeper must be Turned off (Proper classroom decorum [behavior] adopts, Course outlines and calendars explain requirements and assignments, students are responsible for knowing what they say. Students are also responsible for doing all assigned work on time. Excessive absences (more than 04) will result in "F Grade". Students may prepare Sketchbook for taking notes and for references.

Instructor/Tutor

Approved by:

Chairman/ HOD/ Subject Specialist